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# Cosmogenic Activation of Materials using ACTIVIA software

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# Abstract

Cosmogenic activation is a common source of background in ultra-low background, rare event physics. Despite numerous studies being conducted into this field there is not a complete, accurate understanding behind cosmogenic activation. Here, production rates from the ACTIVIA package are presented for elements commonly used in low background experiments. These results are compared using the MENDL-2 and MENDL-2p cross section libraries, and default parameterized cross sections based on semi-empirical formulae developed by Silberberg and Tsao, against available experimental data, as well as previous work using ACTIVIA and MENDL libraries for validity. Results show a good agreement against previous work with ACTIVIA and MENDL, but a poor correlation with experimental data for specific isotopes. It is likely that the poor agreement is a result of a lack of a complete and detailed understanding of the processes involved in cosmogenic activation, as well as using outdated cross section libraries.

## 1 Cosmogenic Activation

The search for rare event physics, such as neutrino-less double-beta decay, and the search for dark matter particles, are highly sensitive to background radiation. Background radioactivity can be provided from activation, primarily from cosmic rays at the surface<sup>[1]</sup>. For all modern rare event experiments this requires underground locations in order to shield from muon and hadronic bombardment. The hadronic component of cosmic rays can be absorbed by a few meters of rock, compared to the kilometers required for muon shielding. Muons produced in the upper atmosphere through cosmic ray interaction are highly penetrating, and can still interact with the target<sup>[2]</sup>. However, the hadronic component of cosmic rays are more highly damaging, and so is the focus of this report.

### 1.1 WIMP detection

Despite the rarity of such events understanding neutrino-less double-beta decay and dark matter physics are fundamental for understanding physics beyond the standard model.

Extra-galactic and galactic observations of astrophysical properties such as rotation curves, and mass-to-light ratios, provide strong evidence for the Universe being comprised of  $\sim 25\%$  dark matter<sup>[3]</sup>. WIMPs (Weakly Interacting Massive Particles) are one of the current favoured candidates. WIMPs incorporate any non-baryonic, massive, non-luminous, and electromagnetically neutral sub-atomic particle. Despite there being no experimental evidence for the existence of WIMPs, several ultra-low background experiments have been undertaken in order to search for them. 'Direct' detection of WIMPs consists of WIMP nucleus scattering events off baryonic matter<sup>[1][4]</sup>. The most critical objective for experimental WIMP scientists is to understand the relationship between the WIMP scattering event rate on a target nucleus, and the recoil energy<sup>[4]</sup>. There are several methods for obtaining a high detection sensitivity for scattering events, including: a high atomic weight ( $A$ ) of the target nucleus, with a high scattering target mass, a low threshold for WIMP scattering, an ultra-low background, and effective background rejection and discrimination techniques<sup>[4]</sup>.

Liquid xenon is one of the target materials used for WIMP searches, often sealed in stainless steel containers<sup>[5][6]</sup>. Xenon is used due to its high scintillation, ionization yield, and radio-purity<sup>[5]</sup>. As there has been no convincing WIMP scattering events, next generation xenon-based detectors are being designed to achieve a detection sensitivity down to a WIMP cross section of  $\sim 10^{-48} \text{ cm}^2$ <sup>[6]</sup>.

### 1.2 Neutrinoless Double-Beta decays

Neutrinoless double-beta decay ( $0\nu\beta\beta$ ) is a weak decay, that occurs using the assumption that neutrinos are Majorana particles. This assumption states that the neutrino and anti-neutrino are indistinguishable, hence, the standard non-exotic weak process of neutrino double-beta decay ( $2\nu\beta\beta$ ) given in Eq. (1), reduces to Eq. (2).<sup>[7]</sup>

$$(A, Z) \rightarrow (A, Z + 2) + e^- + e^- + \bar{\nu}_e + \bar{\nu}_e \quad (1)$$

$$(A, Z) \rightarrow (A, Z + 2) + e^- + e^- \quad (2)$$

It is seen that the  $0\nu\beta\beta$  reaction violates lepton number conservation, due to having a change of +2, hence the neutrinoless decay requires physics beyond the Standard Model<sup>[7]</sup>. The discovery of neutrino oscillations necessitate the need for a neutrino mass, also requiring an extension to the Standard Model. Numerous proposed extensions to the Standard Model do not require lepton number conservation<sup>[2]</sup> and hence could allow for  $0\nu\beta\beta$  decays.

Currently, experimental data is consistent with Majorana and Dirac neutrinos<sup>[2]</sup>, where Dirac neutrinos have distinct particles and anti-particles. Determining the true nature of neutrinos is limited by its mass. However, measurements of the half life of any  $0\nu\beta\beta$  decays can constrain the neutrino mass scale, with the next generation of detectors having sensitivities of 10 – 20 meV<sup>[2]</sup>.

### 1.3 Activation theory

Activation describes the process in which incident radiation induces the production of radioactive isotopes within a material. Cosmic ray activation is a common background source in ultra-low background experiments<sup>[8]</sup>. Activation due to cosmic rays is primarily induced at the surface, with the cosmogenic spectrum being dominated by neutrons, due to the proton energy loss in the atmosphere via ionization. However, at high energies ( $E > 10$  GeV) the spectrum is equivalent for protons and neutrons.

The decay rate of a material ( $\frac{dN}{dt}$ ) is proportional to its production rate ( $R$ ). It also depends on the time exposed to cosmic rays ( $t_{exp}$ ), as well as the time after which the material is not being exposed ( $t_{cool}$ ), often referred to as the cooling time.<sup>[5][8]</sup> The relationship is described in Eq. (3)<sup>[1]</sup>:

$$\frac{dN}{dt} = R(1 - \exp(-\frac{t_{exp}}{\tau})) \exp(-\frac{t_{cool}}{\tau}) \quad (3)$$

The production rate is proportional to the cosmogenic flux received at the Earth's surface ( $\frac{dF}{dE}$ ), and the production cross-section ( $\sigma(E)$ ) for a particular isotope shown in Eq. (4).

$$R = \int \sigma(E) \frac{dF}{dE} dE \quad (4)$$

Experiments are often determining the decay rate of the material given prior knowledge of the production rate. However, it is possible to calculate  $R$  initially if there is knowledge of the production cross-section. In this report  $R$  is what is being determined, using knowledge of  $\sigma(E)$ .

When designing experiments, the commission and engineering runs often take several months to be passed. Along with this, the exact history of the materials used in the commissioned detector, and hence the exact time the materials have been exposed to cosmic rays, is not possible to control. Hence, the value of  $t_{exp}$  is often a large source of uncertainty. Calibration and testing of the equipment, such as photo-multiplier tubes, is carried out underground. While the materials might not be being exposed to cosmic rays, there is still activation contributing to the total radioactivity of the sample. This activation is due to the 'cooling down' time of the material.

The most damaging isotopes in ultra-low background experiments are ones with half lives of  $\sim 1$  month -  $\sim 1$  year. For isotopes with a short half life ( $t_{1/2} < 1$  day) it is possible for the materials to be stored underground and simply wait for the radioactive isotopes produced by activation to decay naturally. Isotopes with a relatively long half life ( $t_{1/2} \gg 1$  year) have a low probability of emission during the calibration, testing, and active phases of the experiment, hence, these isotopes contribute very little to overall background.

Even in perfectly radio-pure materials, cosmogenic activation provides a natural limit to the sensitivity of any experiment<sup>[4]</sup>. In principle cosmogenic activation can be neutralized through processes

such as: minimizing surface exposure, avoiding flights, storing underground, and shielding against cosmic ray bombardment<sup>[8]</sup>. However, these processes usually result in over-complications of the preparation of the experiment. In order to fully understand the effects of activation, all methods of minimization are required to be quantified.

While it is not possible to fully subtract the background while performing ultra-low background experiments, within a certain energy range it is possible to suppress the background contribution. Subtraction is not possible due to the lack of understanding surrounding the background contributors.

In order to account for these isotopes, detailed information of the cross sections and production rates must be known. However, some of the most important activation reactions, such as  $^{60}\text{Co}$  production from heavy elements, have very little experimental data available. As a result of this, a set of semi-empirical formulae based on measurements using incident protons<sup>[10]</sup> was developed by Silberberg and Tsao<sup>[9][10][11]</sup> to calculate the production cross sections. This equation can be used to calculate cross sections for both a proton and a neutron spectra<sup>[11]</sup>. The equation for calculating the partial cross sections is detailed in Eq. (5).

$$\sigma = \sigma_0 f(A) f(E) e^{-P\Delta A} \exp(-R|Z - S + TA^2 + UA^3|^\nu) \Omega \eta \xi \quad (5)$$

Where:  $\sigma_0$  is the cross-section (in millibarns) at energy  $E_0$  ( $\sim 3$  GeV),  $f(A)$  and  $f(E)$  are correction factors applied for heavy elements ( $A \geq 30$ ) or where  $\Delta A \geq 10$ .  $P$  is a function of target mass  $A_t$ , and  $E$ .  $\Delta A$  is the difference between target and product masses (i.e.  $\Delta A = A_t - A$ ).  $R$  represents the width of the cross section,  $S$  describes the peak for various mass numbers,  $T$  describes the increase in neutrons as  $A$  increases,  $U = 3.0 \times 10^{-7}$  and is a correction to improve the agreement with experimental data, and  $\nu$  varies from 1.3 to 2.0 depending on the target and product isotopes.  $\Omega$  represents the nuclear structure factor,  $\eta$  is a nuclear pairing factor (for even and odd  $Z$  and Neutron ( $N$ ) number),  $\xi$  is enhancement factor for light evaporation products, and  $\Omega$ , and  $\eta$  are both of order unity.

## 2 Cosmogenic activation of isotopes at the surface

### 2.1 Code used and input energy spectra

In order to calculate the cosmogenic activation of isotopes the ACTIVIA code was used<sup>[11]</sup>. This allowed for calculations of cosmogenic activation at the surface using an input neutron spectrum, with the Silberberg and Tsao<sup>[9][10][11]</sup> semi-empirical formulae, or tables of cross sections. For this calculation, two neutron spectra were used and their yields compared. The spectra are taken from Gordon et al.<sup>[12]</sup>, and the original ACTIVIA spectrum<sup>[11]</sup> based on parameterization by Armstrong<sup>[13]</sup> and Gerhals<sup>[14]</sup>, shown in Figure 1. It is worth stating that ACTIVIA does not include thermal neutron capture, only fast neutron interaction.<sup>[6][11]</sup>

### 2.2 Cross sections

When calculating the activation yields using ACTIVIA, cross sections for each isotope are also required. Cross sections are the primary source of uncertainty when calculating the yields due to cosmogenic activation, and hence are the major area of study. Cross sections are taken from: the Silberberg and Tsao semi-empirical formulae, which is the default for ACTIVIA, secondly the MENDL-2 library<sup>[15]</sup> for neutron cross sections at energies 0 – 100 MeV, and finally the MENDL-2p library<sup>[16]</sup> for proton cross sections at energies 0 – 200 MeV.

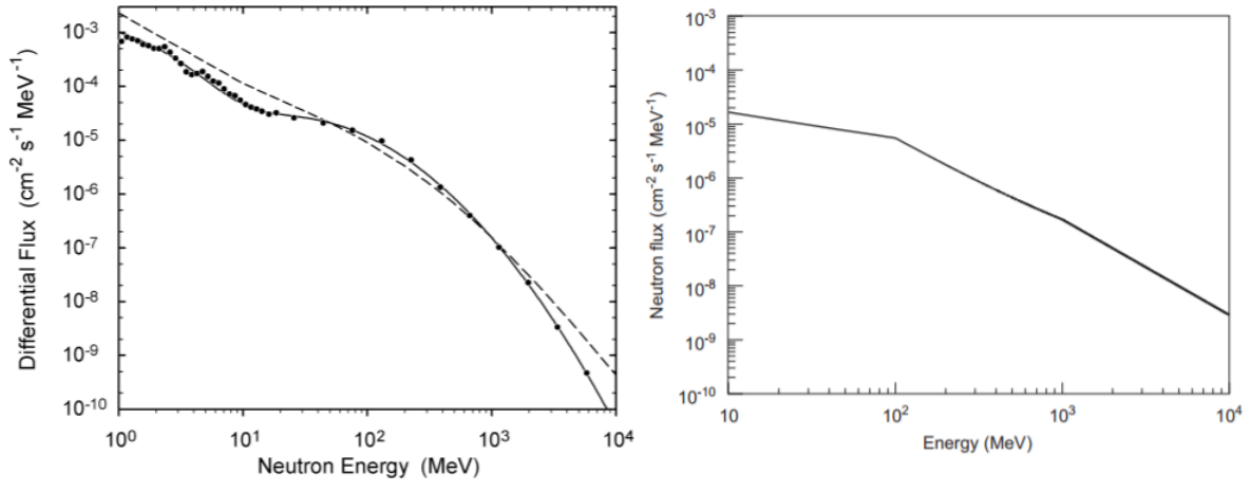


Figure 1: Neutron input spectrum from (left) Gordon et. al.<sup>[12]</sup> spectrum, with the solid curve being the analytical model, data points being the reference spectrum from measurements and the dashed curve being the Ziegel spectrum (see Gordon et al.<sup>[12]</sup> for full reference details), and (right) original neutron spectrum used in ACTIVIA, based on parameterization from Armstrong<sup>[13]</sup> and Gerhals<sup>[14]</sup>.

The uncertainty associated with cross sections is illustrated best by comparison between several different codes available (Fig. 2). The uncertainty in neutron spectra is negligible when compared to cross section uncertainty. This uncertainty is the primary contributor to the lack of a complete and accurate understanding into the field of cosmogenic activation.

It is also worth noting that ACTIVIA uses numerical integration<sup>[11]</sup>, rather than a Monte-Carlo technique to calculate the yields, and so there are no statistical uncertainties associated with the output yields. However, there are still systematic uncertainties arising from using different cross section data.

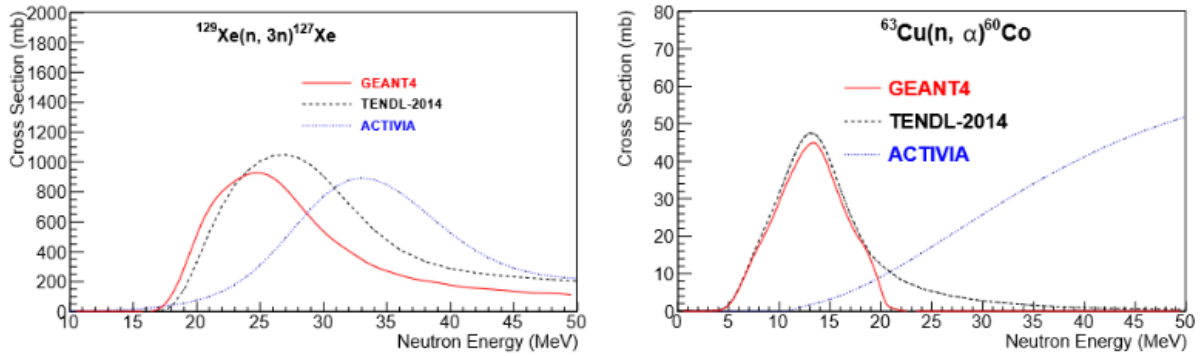


Figure 2: Cross-section comparison between GEANT4<sup>[17]</sup>, TENDL-2014<sup>[18]</sup>, and ACTIVIA parameterization for (left) production of  $^{127}\text{Xe}$  in  $^{129}\text{Xe}$ , and (right)  $^{60}\text{Co}$  production in  $^{63}\text{Cu}$ . Both graphs were taken from Zhang et al.<sup>[6]</sup>

The cross sections taken from MENDL-2<sup>[15]</sup> and MENDL-2p<sup>[16]</sup> that were used by ACTIVIA during calculations have been plotted for  $^{60}\text{Co}$  production in copper, shown in Figure 3. For isotopes that had energy  $> 100$  MeV when running the MENDL-2 libraries, or  $> 200$  MeV when running the MENDL-2p libraries, the data for the cross-sections was taken using the Silberberg and Tsao<sup>[11]</sup> semi-empirical

formulae, which is the default for ACTIVIA<sup>[11]</sup>.

Within the ACTIVIA code there are instructions on how to download and run the MENDL-2p<sup>[16]</sup> libraries, hence why it was used in this project, and is prevalent within the literature.

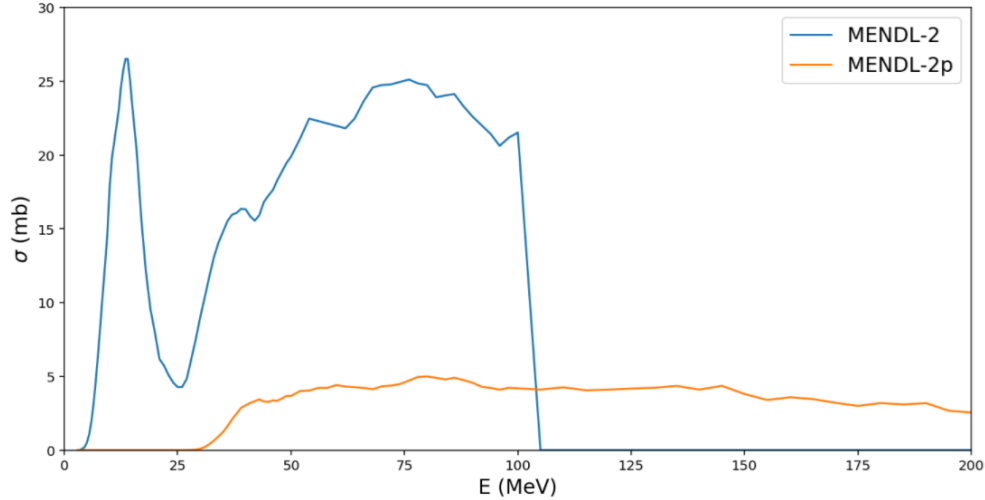


Figure 3: Cross sections for both MENDL-2 and MENDL-2p data for  $^{60}\text{Co}$  production in copper used in ACTIVIA

### 2.3 Target Atoms

For this study the key elements that are associated with ultra-low background experiments were inputted into the ACTIVIA code. The elements included were: argon, copper, germanium, tellurium, titanium, xenon, and stainless steel 316L. As stainless steel is a composite material comprised of numerous elements, each element had to be run through ACTIVIA separately, with the results combined into a single table in accordance with the relative abundances of each element within the material<sup>[19]</sup> shown in Table 1. Each target element also had to have the relative abundance of each isotope included in each input file<sup>[20]</sup>. An example of the input files for each cross section library is given in Figure 4.

Table 1: Relative abundances of elements with stainless steel 316L. Where values had a maximum and a minimum value the average was taken. The remaining component of the material was comprised of iron (Fe), with a relative abundance of 68.545%.

| Grade |     | C    | Mn   | Si   | P     | S    | Cr   | Mo   | Ni   | N    |
|-------|-----|------|------|------|-------|------|------|------|------|------|
| 316L  | Min | -    | -    | -    | -     | -    | 16.0 | 2.00 | 10.0 | -    |
|       | Max | 0.03 | 2.00 | 0.75 | 0.045 | 0.03 | 18.0 | 3.00 | 14.0 | 0.10 |

|                                          |                                          |                                          |
|------------------------------------------|------------------------------------------|------------------------------------------|
| <code>./bin/Activia &lt;&lt; quit</code> | <code>./bin/Activia &lt;&lt; quit</code> | <code>./bin/Activia &lt;&lt; quit</code> |
| 52                                       | 52                                       | 52                                       |
| 8                                        | 8                                        | 8                                        |
| 120 0.0009                               | 120 0.0009                               | 120 0.0009                               |
| 122 0.0255                               | 122 0.0255                               | 122 0.0255                               |
| 123 0.0089                               | 123 0.0089                               | 123 0.0089                               |
| 124 0.0474                               | 124 0.0474                               | 124 0.0474                               |
| 125 0.0707                               | 125 0.0707                               | 125 0.0707                               |
| 126 0.1884                               | 126 0.1884                               | 126 0.1884                               |
| 128 0.3174                               | 128 0.3174                               | 128 0.3174                               |
| 130 0.3408                               | 130 0.3408                               | 130 0.3408                               |
| a                                        | a                                        | a                                        |
| decayData.dat                            | decayData.dat                            | decayData.dat                            |
| 1.0 10000.0 1.0                          | 1.0 10000.0 1.0                          | 1.0 10000.0 1.0                          |
| listOfDataFiles_proton.txt 0.001         | listOfDataFiles.txt 0.001                | 0                                        |
| 1000000000.0 0.0                         | 1000000000.0 0.0                         | 1000000000.0 0.0                         |
| output/Te_All_xSec_n.out 0 0             | output/Te_All_xSec_n.out 0 0             | output/Te_All_xSec_n.out 0 0             |
| output/Te_All_Yields_n.out 0 0           | output/Te_All_Yields_n.out 0 0           | output/Te_All_Yields_n.out 0 0           |
| quit                                     | quit                                     | quit                                     |

Figure 4: An example of the input file used by ACTIVIA, this example is for tellurium. The format of the file is: proton number (Z), number of isotopes, atomic weight (A) of each product, abundance of each product, 'a' is for all isotopes, decay model, start from 1 MeV - 10000 MeV in 1 MeV intervals, cross section data (cross section data is shown by *(left)* 0, which is for using the Silberberg and Tsao<sup>[11]</sup> cross sections, *(middle)* MENDL-2<sup>[15]</sup> libraries, and *(right)* MENDL-2p<sup>[16]</sup> libraries), exposed for  $1 \times 10^9$  days with 0 cooling time, and finally the output names.

Within the input files the energy range (in MeV), exposure times and cooling times (both given in days) had to be specified for each target element. For our calculations, the energy range was set to be from 1.0 MeV - 10,000 MeV with steps of 1.0 MeV, with an exposure time of  $t_{exp} = 10^9$  days, and a cooling time of  $t_{cool} = 0.0$  days. As the exposure time was significantly long, it can be assumed that it is infinite, to ensure the decay rate is equal to the production rate shown in Eq. (3). This allows for the system to be in equilibrium, and so can be compared to experimental data, and previous calculations, without having to correct for exposure, and 'cooling down' time.

### 3 Cosmogenic activation rates in targeted atoms

The activation rates calculated by ACTIVIA for copper, and germanium are detailed in Table 2, and Table 3. The complete data set for all targeted isotopes is given in the appendix.

It is observed that for all targeted atoms the lightest isotopes ( $^3\text{H}$ ,  $^7\text{Be}$ ,  $^{11}\text{B}$  etc.) produce the same output across all cross sections when calculating with the same spectra. This is likely a result of the limited energy range of the MENDL-2 and MENDL-2p libraries. Hence, these isotopes are produced outside the energy range of the MENDL-2 and MENDL-2p libraries, and so the ACTIVIA code defaults to using the Silberberg and Tsao<sup>[10][11]</sup> semi-empirical formula for cross sections. This results in the production rates being calculated as the same value as seen in Table 2, and Table 3.

Table 2: Production rates of radioactive isotopes in copper in units of  $\text{kg}^{-1} \text{ day}^{-1}$ , using the original parameterized cross sections based on Silberberg and Tsao<sup>[10]</sup>, MENDL-2p<sup>[15]</sup>, and MENDL-2<sup>[16]</sup>. Each of the cross-section data were ran through ACTIVIA<sup>[11]</sup> using the spectra based on parameterization from Armstrong<sup>[13]</sup> and Gerhals<sup>[14]</sup>, and the Gordon et al. neutron spectra<sup>[12]</sup>.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |         |               |         |              |         |
|---------|---------------------|--------------------------------------------------------|---------|---------------|---------|--------------|---------|
|         |                     | Semi-empirical formulae [9][10]                        |         | MENDL-2p [15] |         | MENDL-2 [16] |         |
|         |                     | [11]                                                   | [12]    | [11]          | [12]    | [11]         | [12]    |
| H-3     | 4503.4              | 35.880                                                 | 50.704  | 35.880        | 50.704  | 35.880       | 50.704  |
| Be-7    | 53.3                | 1.078                                                  | 0.543   | 1.078         | 0.543   | 1.078        | 0.543   |
| Be-10   | 5.84E+08            | 3.612                                                  | 7.782   | 3.612         | 7.782   | 3.612        | 7.782   |
| B-11    | 1.10E+10            | 0.896                                                  | 1.938   | 0.896         | 1.938   | 0.896        | 1.938   |
| C-14    | 2.09E+06            | 0.478                                                  | 0.514   | 0.478         | 0.514   | 0.478        | 0.514   |
| Ne-20   | 1.10E+10            | 0.030                                                  | 0.016   | 0.030         | 0.016   | 0.030        | 0.016   |
| Na-22   | 950.35              | 0.311                                                  | 0.160   | 0.311         | 0.160   | 0.311        | 0.160   |
| Al-26   | 2.66E+08            | 0.213                                                  | 0.108   | 0.213         | 0.108   | 0.213        | 0.108   |
| Si-32   | 237000              | 0.133                                                  | 0.077   | 0.133         | 0.077   | 0.133        | 0.077   |
| P-31    | 1.10E+10            | 0.004                                                  | 0.025   | 0.045         | 0.025   | 0.045        | 0.025   |
| P-32    | 14.28               | 0.834                                                  | 0.484   | 0.834         | 0.484   | 0.834        | 0.484   |
| P-33    | 25.3                | 0.449                                                  | 0.269   | 0.449         | 0.269   | 0.449        | 0.269   |
| S-35    | 87.4                | 0.705                                                  | 0.459   | 0.705         | 0.459   | 0.705        | 0.459   |
| Cl-36   | 1.10E+08            | 1.293                                                  | 0.885   | 1.293         | 0.885   | 1.293        | 0.885   |
| Ar-37   | 35                  | 0.705                                                  | 0.508   | 0.705         | 0.508   | 0.705        | 0.508   |
| Ar-39   | 98200               | 1.195                                                  | 0.927   | 1.195         | 0.927   | 1.195        | 0.927   |
| Ar-42   | 12000               | 0.039                                                  | 0.035   | 0.039         | 0.035   | 0.039        | 0.035   |
| K-42    | 0.52                | 0.585                                                  | 0.524   | 0.585         | 0.524   | 0.585        | 0.524   |
| K-43    | 0.93                | 0.208                                                  | 0.196   | 0.208         | 0.196   | 0.208        | 0.196   |
| Ca-40   | 1.10E+10            | 0.012                                                  | 0.010   | 0.012         | 0.010   | 0.012        | 0.010   |
| Ca-41   | 3.65E+07            | 0.721                                                  | 0.626   | 0.721         | 0.626   | 0.721        | 0.626   |
| Ca-45   | 165                 | 0.328                                                  | 0.349   | 0.328         | 0.349   | 0.328        | 0.349   |
| Ca-47   | 4.536               | 0.026                                                  | 0.032   | 0.026         | 0.032   | 0.026        | 0.032   |
| Sc-46   | 83.8                | 3.132                                                  | 3.551   | 3.132         | 3.551   | 3.114        | 3.512   |
| Sc-47   | 3.42                | 0.620                                                  | 0.746   | 0.620         | 0.746   | 0.614        | 0.731   |
| Sc-48   | 1.82                | 0.167                                                  | 0.213   | 0.167         | 0.213   | 0.166        | 0.212   |
| Ti-44   | 17200               | 0.163                                                  | 0.168   | 0.166         | 0.176   | 0.163        | 0.168   |
| V-48    | 15.976              | 2.946                                                  | 3.880   | 3.000         | 4.074   | 2.909        | 3.798   |
| V-49    | 330                 | 8.113                                                  | 11.331  | 7.601         | 10.209  | 7.918        | 10.884  |
| V-51    | 1.10E+10            | 0.132                                                  | 0.205   | 0.111         | 0.155   | 0.126        | 0.191   |
| Cr-48   | 0.989               | 0.144                                                  | 0.191   | 0.169         | 0.257   | 0.144        | 0.191   |
| Cr-51   | 27.7                | 9.996                                                  | 15.850  | 9.721         | 15.398  | 9.643        | 15.040  |
| Mn-52   | 5.59                | 4.101                                                  | 6.927   | 5.409         | 10.259  | 4.027        | 6.739   |
| Mn-53   | 1.38E+09            | 5.849                                                  | 10.417  | 6.132         | 11.246  | 5.580        | 9.785   |
| Mn-54   | 312                 | 14.319                                                 | 26.575  | 12.470        | 22.284  | 13.798       | 25.296  |
| Fe-55   | 986.2               | 19.321                                                 | 38.148  | 24.406        | 50.434  | 20.934       | 41.677  |
| Fe-59   | 44.6                | 4.239                                                  | 9.716   | 1.730         | 3.616   | 2.930        | 6.413   |
| Fe-60   | 5.47E+08            | 0.577                                                  | 1.306   | 0.201         | 0.380   | 0.429        | 0.936   |
| Co-56   | 78.8                | 8.736                                                  | 18.081  | 14.122        | 31.172  | 9.972        | 21.070  |
| Co-57   | 271                 | 32.443                                                 | 70.216  | 36.433        | 80.854  | 35.814       | 79.173  |
| Co-58   | 70.8                | 56.611                                                 | 126.538 | 38.127        | 81.837  | 55.029       | 123.819 |
| Co-60   | 1925.2              | 26.279                                                 | 62.417  | 9.675         | 21.077  | 23.251       | 54.725  |
| Ni-56   | 6.1                 | 0.170                                                  | 0.351   | 0.585         | 1.371   | 0.156        | 0.319   |
| Ni-57   | 1.5                 | 1.497                                                  | 3.239   | 4.572         | 10.875  | 1.501        | 3.221   |
| Ni-59   | 2.92E+07            | 51.472                                                 | 119.638 | 62.343        | 146.581 | 56.437       | 131.071 |
| Ni-63   | 36524               | 63.226                                                 | 177.720 | 43.929        | 128.621 | 49.408       | 152.503 |
| Cu-63   | 1.10E+10            | 1.422                                                  | 3.416   | 2.433         | 6.252   | 3.098        | 8.067   |
| Zn-65   | 244.1               | 19.581                                                 | 61.215  | 14.619        | 86.520  | 19.581       | 61.215  |



Table 3: Production rates of radioactive isotopes in germanium in units of  $\text{kg}^{-1} \text{ day}^{-1}$ . See Table 2 for details on each column and row.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |         |               |         |              |         |
|---------|---------------------|--------------------------------------------------------|---------|---------------|---------|--------------|---------|
|         |                     | Semi-empirical formulae [9][10]                        |         | MENDL-2p [15] |         | MENDL-2 [16] |         |
|         |                     | [11]                                                   | [12]    | [11]          | [12]    | [11]         | [12]    |
| H-3     | 4503.4              | 34.118                                                 | 45.996  | 34.118        | 45.996  | 34.118       | 45.996  |
| Be-7    | 53.3                | 0.899                                                  | 0.409   | 0.899         | 0.409   | 0.899        | 0.409   |
| Be-10   | 5.84E+08            | 3.696                                                  | 7.997   | 3.696         | 7.997   | 3.696        | 7.997   |
| B-11    | 1.10E+10            | 0.051                                                  | 0.013   | 0.051         | 0.013   | 0.051        | 0.013   |
| C-14    | 2.09E+06            | 0.590                                                  | 0.152   | 0.590         | 0.152   | 0.590        | 0.152   |
| Ne-20   | 1.10E+10            | 0.049                                                  | 0.009   | 0.049         | 0.009   | 0.049        | 0.009   |
| Na-22   | 950.35              | 0.194                                                  | 0.065   | 0.194         | 0.065   | 0.194        | 0.065   |
| Al-26   | 2.66E+08            | 0.130                                                  | 0.044   | 0.130         | 0.044   | 0.130        | 0.044   |
| Si-32   | 237000              | 0.053                                                  | 0.021   | 0.053         | 0.021   | 0.053        | 0.021   |
| P-31    | 1.10E+10            | 0.022                                                  | 0.009   | 0.022         | 0.009   | 0.022        | 0.009   |
| P-32    | 14.28               | 0.399                                                  | 0.168   | 0.399         | 0.168   | 0.399        | 0.168   |
| P-33    | 25.3                | 0.211                                                  | 0.088   | 0.211         | 0.088   | 0.211        | 0.088   |
| S-35    | 87.4                | 0.297                                                  | 0.147   | 0.297         | 0.147   | 0.297        | 0.147   |
| Cl-36   | 1.10E+08            | 0.614                                                  | 0.313   | 0.614         | 0.313   | 0.614        | 0.313   |
| Ar-37   | 35                  | 0.362                                                  | 0.194   | 0.362         | 0.194   | 0.362        | 0.194   |
| Ar-39   | 98200               | 0.557                                                  | 0.306   | 0.557         | 0.306   | 0.557        | 0.306   |
| Ar-42   | 12000               | 0.018                                                  | 0.011   | 0.018         | 0.011   | 0.018        | 0.011   |
| K-42    | 0.52                | 0.260                                                  | 0.156   | 0.260         | 0.156   | 0.260        | 0.156   |
| K-43    | 0.93                | 0.092                                                  | 0.057   | 0.092         | 0.057   | 0.092        | 0.057   |
| Ca-40   | 1.10E+10            | 0.006                                                  | 0.004   | 0.006         | 0.004   | 0.006        | 0.004   |
| Ca-41   | 3.65E+07            | 0.337                                                  | 0.205   | 0.337         | 0.205   | 0.337        | 0.205   |
| Ca-45   | 165                 | 0.138                                                  | 0.091   | 0.138         | 0.091   | 0.138        | 0.091   |
| Ca-47   | 4.536               | 0.011                                                  | 0.007   | 0.011         | 0.007   | 0.011        | 0.007   |
| Sc-46   | 83.8                | 0.605                                                  | 0.416   | 0.605         | 0.416   | 0.605        | 0.416   |
| Sc-47   | 3.42                | 0.243                                                  | 0.171   | 0.243         | 0.171   | 0.243        | 0.171   |
| Sc-48   | 1.82                | 0.062                                                  | 0.045   | 0.062         | 0.045   | 0.062        | 0.045   |
| Ti-44   | 17200               | 0.081                                                  | 0.055   | 0.081         | 0.055   | 0.081        | 0.055   |
| V-48    | 15.976              | 1.042                                                  | 0.807   | 1.042         | 0.807   | 1.042        | 0.807   |
| V-49    | 330                 | 2.385                                                  | 1.885   | 2.385         | 1.885   | 2.385        | 1.885   |
| V-51    | 1.10E+10            | 0.038                                                  | 0.031   | 0.038         | 0.031   | 0.038        | 0.031   |
| Cr-48   | 0.989               | 0.063                                                  | 0.050   | 0.063         | 0.050   | 0.063        | 0.050   |
| Cr-51   | 27.7                | 2.471                                                  | 2.147   | 2.477         | 2.162   | 2.471        | 2.147   |
| Mn-52   | 5.59                | 1.037                                                  | 0.967   | 1.059         | 1.024   | 1.037        | 0.967   |
| Mn-53   | 1.38E+09            | 1.186                                                  | 1.142   | 1.210         | 1.207   | 1.186        | 1.142   |
| Mn-54   | 312                 | 2.528                                                  | 2.456   | 2.573         | 2.574   | 2.528        | 2.456   |
| Fe-55   | 986.2               | 3.265                                                  | 3.534   | 3.562         | 4.304   | 3.265        | 3.533   |
| Fe-59   | 44.6                | 0.347                                                  | 0.431   | 0.343         | 0.419   | 0.350        | 0.436   |
| Fe-60   | 5.47E+08            | 0.116                                                  | 0.152   | 0.111         | 0.139   | 0.116        | 0.153   |
| Co-56   | 78.8                | 1.788                                                  | 2.083   | 2.054         | 2.764   | 1.788        | 2.083   |
| Co-57   | 271                 | 6.337                                                  | 7.716   | 6.860         | 9.055   | 6.326        | 7.692   |
| Co-58   | 70.8                | 7.958                                                  | 9.866   | 8.360         | 10.881  | 7.974        | 9.899   |
| Co-60   | 1925.2              | 2.662                                                  | 3.542   | 2.699         | 3.636   | 2.704        | 3.641   |
| Ni-56   | 6.1                 | 0.035                                                  | 0.043   | 0.055         | 0.095   | 0.035        | 0.043   |
| Ni-57   | 1.5                 | 0.224                                                  | 0.284   | 0.395         | 0.722   | 0.224        | 0.283   |
| Ni-59   | 2.92E+07            | 3.831                                                  | 5.333   | 6.156         | 11.095  | 3.937        | 5.577   |
| Ni-63   | 36524               | 1.409                                                  | 2.276   | 1.550         | 2.614   | 1.727        | 3.048   |
| Ni-66   | 2.28                | 0.061                                                  | 0.104   | 0.053         | 0.085   | 0.085        | 0.162   |
| Cu-63   | 1.10E+10            | 0.678                                                  | 1.138   | 1.152         | 2.294   | 0.900        | 1.671   |
| Cu-67   | 2.58                | 0.333                                                  | 0.560   | 0.595         | 1.195   | 1.113        | 2.492   |
| Zn-65   | 244.1               | 19.369                                                 | 38.910  | 29.145        | 62.825  | 24.996       | 52.851  |
| Zn-72   | 1.938               | 0.877                                                  | 1.505   | 0.716         | 1.105   | 1.875        | 4.066   |
| Ga-67   | 3.263               | 43.432                                                 | 95.500  | 45.703        | 101.913 | 47.763       | 106.844 |
| Ga-69   | 1.10E+10            | 3.388                                                  | 7.659   | 2.243         | 4.798   | 4.051        | 9.369   |
| Ga-71   | 0.59                | 52.087                                                 | 135.416 | 27.765        | 73.613  | 33.894       | 80.812  |
| Ge-68   | 270.8               | 10.242                                                 | 22.659  | 44.462        | 112.549 | 34.165       | 87.290  |
| Ge-69   | 1.625               | 31.576                                                 | 76.350  | 59.017        | 147.526 | 58.772       | 147.442 |
| Ge-71   | 11.2                | 74.439                                                 | 186.416 | 95.516        | 241.393 | 126.225      | 323.402 |
| As-72   | 1.083               | 49.465                                                 | 138.576 | 51.905        | 160.593 | 49.465       | 138.576 |
| As-73   | 80.3                | 39.883                                                 | 108.397 | 64.512        | 179.569 | 39.883       | 108.397 |
| As-74   | 17.8                | 27.235                                                 | 82.793  | 29.443        | 121.649 | 27.235       | 82.793  |
| As-76   | 1.096               | 4.470                                                  | 13.973  | 2.766         | 17.148  | 4.470        | 13.973  |

### 3.1 Comparison between calculations and experimental data

In order to test the reliability of the ACTIVIA code the output yields are compared for both input spectra against the default parameterized cross sections, MENDL-2, and MENDL-2p libraries, with other literature outputs also using MENDL-2p cross sections in ACTIVIA.

Table 4: Comparison of production rates of radioactive isotopes in copper computed by ACTIVIA using the semi-empirical formulae<sup>[9][10]</sup> and MENDL-2p<sup>[16]</sup> libraries for cross sections. 'This work' refers to outputs calculated during this project, with [5] referring to calculations from Baudis et al.<sup>[5]</sup>. The default ACTIVIA spectrum<sup>[11]</sup> was used for both calculations.

| Isotope | Half-life(days) | Production rates in copper ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |          |                                            |          |
|---------|-----------------|------------------------------------------------------------------|----------|--------------------------------------------|----------|
|         |                 | This work                                                        |          | [5]                                        |          |
|         |                 | Semi-empirical formulae <sup>[9][10]</sup>                       | MENDL-2p | Semi-empirical formulae <sup>[9][10]</sup> | MENDL-2p |
| Sc-46   | 83.79           | 3.13                                                             | 3.13     | 3.11                                       | 3.11     |
| V-48    | 15.97           | 2.95                                                             | 3.00     | -                                          | -        |
| Mn-54   | 312.12          | 14.3                                                             | 12.5     | 14.3                                       | 12.5     |
| Fe-59   | 44.5            | 4.24                                                             | 1.73     | 4.23                                       | 1.81     |
| Co-56   | 77.24           | 8.74                                                             | 14.1     | 8.73                                       | 14.1     |
| Co-57   | 271.74          | 32.4                                                             | 36.4     | 32.5                                       | 36.4     |
| Co-58   | 70.86           | 56.6                                                             | 38.1     | 56.6                                       | 38.1     |
| Co-60   | 1925.28         | 26.3                                                             | 9.67     | 26.3                                       | 9.68     |

Table 5: Comparisons of production rates of radioactive isotopes in copper for the default ACTIVIA spectrum<sup>[11]</sup> based on parameterization<sup>[13][14]</sup>, and the Gordon et al.<sup>[12]</sup> spectrum. Cross section data is only taken from the Silberberg and Tsao semi-empirical formulae<sup>[9][10]</sup>. The values from this project labelled 'This work' are compared against literature values originally taken from Zhang et al.<sup>[6]</sup>

| Isotope | Half-life(days) | Production rates in copper ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |      |       |
|---------|-----------------|------------------------------------------------------------------|-------|------|-------|
|         |                 | This work                                                        |       | [6]  |       |
|         |                 | [11]                                                             | [12]  | [11] | [12]  |
| Sc-46   | 83.79           | 3.13                                                             | 3.55  | 3.13 | 4.09  |
| V-48    | 15.97           | 2.95                                                             | 3.88  | -    | -     |
| Mn-54   | 312.12          | 14.3                                                             | 26.6  | 14.3 | 30.0  |
| Fe-59   | 44.5            | 4.24                                                             | 9.72  | 4.24 | 10.50 |
| Co-56   | 77.24           | 8.74                                                             | 18.1  | 8.7  | 20.1  |
| Co-57   | 271.74          | 32.4                                                             | 70.2  | 32.4 | 77.5  |
| Co-58   | 70.86           | 56.6                                                             | 126.5 | 56.6 | 138.1 |
| Co-60   | 1925            | 26.3                                                             | 62.4  | 26.3 | 66.1  |

In addition to this the yields of specific isotopes of copper were compared against available experimental data from Baudis et. al<sup>[5]</sup>, and the LNGS<sup>[21]</sup>, as well as calculations using the Cosmo<sup>[22]</sup> code, shown in Table 6. As the values taken from Baudis et. al, Cosmo, and the LNGS were in units of  $\mu\text{Bq kg}^{-1}$ , and outputs calculated with ACTIVIA were in units of  $\text{kg}^{-1} \text{ day}^{-1}$ , the literature values were multiplied by 0.0864 to convert into comparable units.

Table 6: Production rates of radioactive isotopes in copper produced in ACTIVIA compared against experimental measurements from Baudis et al.<sup>[5]</sup>, and LNGS<sup>[19]</sup>, and the calculation using the Cosmo<sup>[20]</sup> code.

| Isotope | Half-life(days) | Production rates in copper (kg <sup>-1</sup> day <sup>-1</sup> ) |        |          |       |         |        |                   |             |      | Cosmo [22] |
|---------|-----------------|------------------------------------------------------------------|--------|----------|-------|---------|--------|-------------------|-------------|------|------------|
|         |                 | This work                                                        |        |          |       |         |        | Literature values |             |      |            |
|         |                 | Semi-empirical formulae [5][6]                                   |        | MENDL-2p |       | MENDL-2 |        | Experiments       |             |      |            |
|         |                 | [11]                                                             | [12]   | [11]     | [12]  | [11]    | [12]   | Baudis et al. [5] | LNGS [21]   |      |            |
| Sc-46   | 83.79           | 3.13                                                             | 3.55   | 3.13     | 3.55  | 3.11    | 3.51   | 2.33(+0.95 -0.78) | 2.18 ± 0.74 | 1.47 |            |
| V-48    | 15.97           | 2.95                                                             | 3.88   | 3.00     | 4.07  | 2.91    | 3.80   | 3.37(+1.64 -1.30) | 4.49 ± 1.64 | 3.11 |            |
| Mn-54   | 312.12          | 14.32                                                            | 26.57  | 12.47    | 22.28 | 13.80   | 25.30  | 13.3(+3.02 -2.94) | 34.0 ± 3.37 | 13.5 |            |
| Fe-59   | 44.5            | 4.24                                                             | 9.72   | 1.73     | 3.62  | 2.93    | 6.41   | 4.06(+1.38 -1.21) | 4.92 ± 1.21 | 4.32 |            |
| Co-56   | 77.24           | 8.74                                                             | 18.08  | 14.12    | 31.17 | 9.97    | 21.07  | 9.33(+1.21 -1.38) | 9.50 ± 1.21 | 7    |            |
| Co-57   | 271.74          | 32.44                                                            | 70.22  | 36.43    | 80.85 | 35.81   | 79.17  | 44.8(+8.64 -8.21) | 74.3 ± 16.4 | 30.2 |            |
| Co-58   | 70.86           | 56.61                                                            | 126.54 | 38.13    | 81.84 | 55.03   | 123.82 | 68.9(+5.36 -5.01) | 67.9 ± 3.72 | 54.6 |            |
| Co-60   | 1925.28         | 26.28                                                            | 62.42  | 9.67     | 21.08 | 23.25   | 54.72  | 29.4(+7.08 -5.88) | 86.4 ± 7.78 | 25.7 |            |

Due to xenon’s importance in the field of ultra-low background experiments, production rates were also compared against literature values for xenon, shown in Table 7.

Table 7: Production rates of radioactive isotopes in xenon calculated by ACTIVIA, compared against experimental data taken from Baudis et al.<sup>[5]</sup>, and calculations done using the Cosmo<sup>[20]</sup> code. The literature values for Cosmo use a different input spectrum to the ACTIVIA spectrum ([11]), while the Baudis et al.<sup>[5]</sup>, and LNGS<sup>[19]</sup> measurements are based on experiments.

| Isotope | Half-life (days) | Production rates in xenon ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |          |       |                   |       |                    |             |
|---------|------------------|-----------------------------------------------------------------|-------|----------|-------|-------------------|-------|--------------------|-------------|
|         |                  | This work                                                       |       |          |       | Literature values |       |                    |             |
|         |                  | Semi-empirical formulae [5][6]                                  |       | MENDL-2p |       | MENDL-2           |       | Experiments        | Calculation |
|         |                  | [11]                                                            | [12]  | [11]     | [12]  | [11]              | [12]  | Baudis et. Al [5]  | Cosmo [20]  |
| Be-7    | 53.3             | 0.548                                                           | 0.209 | 0.548    | 0.209 | 0.548             | 0.209 | 32.0 (+20.7 -19.9) | 0.553       |
| Rh-101  | 1205             | 1.43                                                            | 0.993 | 1.43     | 0.993 | 1.43              | 0.993 | 123(+83.8 -73.4)   | 1.32        |
| Sb-125  | 986              | 0.016                                                           | 0.007 | 0.025    | 0.028 | 0.046             | 0.079 | 51.0(+22.5 -19.9)  | 1.67        |
| I-126   | 13               | 21.3                                                            | 48.3  | 9.51     | 19.1  | 19.7              | 44.2  | 15.1(+8.12 -7.52)  | 21.3        |
| Xe-127  | 36.4             | 35.7                                                            | 89.6  | 66.5     | 168   | 62.3              | 160   | 162(+25.1 -23.3)   | 48.0        |

### 3.2 Discussion of results

In general, comparison between the work completed in this project show good agreement with previous work using the ACTIVIA code, and MENDL-2p libraries. Table 4 shows that for all isotopes (excluding <sup>48</sup>V where no data was present in Baudis et al.<sup>[5]</sup>) the production rates calculated in this project and in Baudis et al.<sup>[5]</sup> are comparable, when using the default input spectrum in ACTIVIA<sup>[11]</sup>. This was done in order to validate the correct usage of the code.

However, Table 5 does show that while the production rates calculated in this work and Zhang et al.<sup>[6]</sup> are identical (to 3 sf) when using the default ACTIVIA spectrum<sup>[11]</sup>, rates calculated using the Gordon et al.<sup>[12]</sup> spectrum differ by an average of  $\sim 10\%$  across all isotopes shown. Both calculations involve the same input neutron spectrum from Gordon et al.<sup>[12]</sup>, with the same version of the ACTIVIA code. However, in Zhang et al.<sup>[6]</sup> they use data points for the cross sections, whereas we are using the semi-empirical formulae by Silberberg and Tsao<sup>[10]</sup>, which could explain the slight difference in production rates. Although, as the discrepancy is relatively insignificant, it is still possible to say that the rates calculated using the Gordon et al.<sup>[12]</sup> input spectra agree well with the rates calculated in Zhang et al.<sup>[6]</sup>.

While the results for the semi-empirical cross sections, and MENDL-2p cross sections agree well with values found in the literature, there is very little evidence of values calculated with MENDL-2<sup>[16]</sup> neutron cross sections. This is likely due to the limited energy range ( $0 - 100 \text{ MeV}$ <sup>[16]</sup>) of the MENDL-2

libraries making them unsuitable for calculations. However, one of the aims of this project was test new cross sections

The MENDL libraries do however, have a very limited energy range that restricts their usefulness in calculations of activation. MENDL-2 libraries only have an energy range of 0 – 100 MeV<sup>[15]</sup>, with MENDL-2p libraries having an energy range of 0 – 200 MeV<sup>[16]</sup>. Hence, as calculations were performed from 1 MeV to 10 GeV, the majority of calculations were using the Silberberg and Tsao semi-empirical formulae<sup>[10]</sup> for cross section data. It should also be noted that the MENDL libraries were last updated in 2000<sup>[15]</sup> and 2001<sup>[16]</sup> for MENDL-2 and MENDL-2p respectively. Therefore, it is not unreasonable to assume that these cross section libraries are outdated, increasing the uncertainty in the validity of our results. However, the Silberberg and Tsao formulae for cross sections are even older, with them last being updated in 1994. Hence, if we were to repeat these calculations with updated cross sections, we would have to use cross section libraries from places like the TENDL-2014<sup>[18]</sup> data base.

However, comparisons between the work produced in this project using ACTIVIA, and results from experimental values by Baudis et al.<sup>[5]</sup>, and the LNGS<sup>[21]</sup> show little agreement. Table 6 shows that there is only agreement between the Baudis et al.<sup>[5]</sup> values for <sup>46</sup>Sc, <sup>48</sup>V, <sup>59</sup>Fe, and <sup>54</sup>Mn (only when using the original ACTIVIA spectrum<sup>[11]</sup> for <sup>54</sup>Mn), and the LNGS<sup>[21]</sup> values for <sup>46</sup>Sc, <sup>48</sup>V, and <sup>59</sup>Fe. Table 7 shows an almost complete lack of agreement between calculated values of production rates for xenon in ACTIVIA compared with the Baudis et. al<sup>[5]</sup> values.

Yet despite the lack of overall agreement between the rates calculated by ACTIVIA to the experimental values, it should be noted that there is also a lack of consistency within the experiments themselves. 3 of the 8 total radioactive isotopes produced in copper lie outside the range of errors for Baudis et al.<sup>[5]</sup> and LNGS<sup>[21]</sup> measurements. For <sup>54</sup>Mn, <sup>57</sup>Co, and <sup>60</sup>Co the rates measured by Baudis et al<sup>[5]</sup> are a factor of  $\sim 2$  lower than those measured by LNGS<sup>[21]</sup>. This highlights the lack of a clear and accurate understanding behind the processes which lead to activation. Therefore, it is difficult to make any meaningful comparisons between experimental values, and the ACTIVIA values calculated, due to this lack of consistency within experiments.

Comparisons can be made between the Cosmo<sup>[22]</sup> code and the work produced in this project using ACTIVIA. Tables 6 and 7 show that the Cosmo<sup>[22]</sup> code shows some agreement with the results calculated using ACTIVIA, particularly for results in ACTIVIA using the default parameterized spectrum<sup>[11]</sup>. This is likely due to the fact that the Cosmo code was first produced in 1992<sup>[22]</sup>, and uses an approximate input spectrum based on work from Lal and Peters<sup>[23]</sup>, which was first developed in 1967<sup>[23]</sup>. Cosmo also uses the semi-empirical formulae developed by Silberberg and Tsao<sup>[10]</sup> for its cross sections, however, as the code was produced in 1992<sup>[22]</sup>, and the last updates to the formulae being in 1994, the Cosmo code will not have the most recent version of the semi-empirical formulae. The values calculated using the Gordon et al<sup>[12]</sup> input spectrum are not comparable to the Cosmo values for the majority of the radioactive isotopes compared in Tables 6 and 7. As the Gordon et al.<sup>[12]</sup> spectra is more modern, and based on measurements of the neutron flux, it is likely that the Gordon et al.<sup>[12]</sup> spectrum represents a more accurate input spectrum than the default ACTIVIA spectrum<sup>[11]</sup>.

It was worth noting that despite the Cosmo code being older than the ACTIVIA code, it is unlikely that the code structure produced such vastly different values for several of the radioactive isotopes compared. The input spectrum, and cross sections used, are significantly more crucial, and represent the major source of the discrepancy.

## 4 Conclusion

We have presented data of production of radioactive isotopes by cosmogenic activation, in a number of materials that are important when considering ultra-low background experiments. The ultra-low background experiments test physics beyond the standard model, such as: WIMP scattering events, and neutrinoless double-beta decay.

Calculations for the production rate were performed using the ACTIVIA code structure<sup>[11]</sup>, using the default input spectrum based on parameterization<sup>[13][14]</sup>, and the Gordon et al.<sup>[12]</sup> neutron spec-

trum. ACTIVIA also required the use of data for cross-sections for each of the target atoms described in section 2.3. These cross-sections were taken from: semi-empirical formulae developed by Silberberg and Tsao<sup>[10]</sup>, and the MENDL-2<sup>[15]</sup> and MENDL-2p<sup>[16]</sup> libraries.

The results for each element, ran through ACTIVIA with both input spectra, and all three cross section sources, were tabulated, with specific isotopes of copper and xenon compared against existing data to check the validity of the tests.

Comparisons show that the work produced in ACTIVIA with MENDL cross sections agrees very well with literature data produced in the same manner (Table 4 and 5). However, when comparing ACTIVIA results to experimental measurements, and calculations using other codes, of production rates of radioactive isotopes in copper and xenon show little agreement for the majority of produced radioactive isotopes.

However, despite the lack of agreement to the experimental values with ACTIVIA, the fact that the experimental values are not consistent with each other mean that it is difficult to draw any meaningful conclusions on the validity of the calculations in ACTIVIA. This reinforces the notion that there is a lack of a complete, and accurate, understanding of the full processes that lead to cosmogenic activation.

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## 7 Appendix

The complete and full data tables of the targeted materials used in the ACTIVIA calculations are shown below.

Initially shown are the individual elements that were ran through ACTIVIA, then afterwards are the constituents of stainless steel that also had to be run through ACTIVIA individually.

Table 8: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in argon by ACTIVIA. Cross section data for the calculations is taken using the original parameterized cross sections based on the semi-empirical formulae developed by Silberberg and Tsao<sup>[10]</sup>, the MENDL-2p<sup>[15]</sup> proton library, and the MENDL-2<sup>[16]</sup> neutron library. Two separate input spectra are used to calculate the production rates. [11] refers to the default parameterized spectra<sup>[11]</sup> used in ACTIVIA, based on parameterization from Armstrong<sup>[13]</sup> and Gerhals<sup>[14]</sup>, and [12] refers to the Gordon et al.<sup>[12]</sup> neutron spectra.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae<br>[9][10]                     |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 45.60                                                  | 74.05  | 45.58         | 74.05  | 45.58        | 74.05  |
| Be-7    | 53.3                | 1.77                                                   | 1.21   | 1.77          | 1.21   | 1.77         | 1.21   |
| Be-10   | 5.84E+08            | 1.37                                                   | 0.98   | 1.37          | 0.98   | 1.37         | 0.98   |
| B-11    | 1.10E+10            | 0.60                                                   | 1.34   | 0.60          | 1.34   | 0.60         | 1.34   |
| C-14    | 2.09E+06            | 4.32                                                   | 8.67   | 4.32          | 8.67   | 4.32         | 8.67   |
| Ne-20   | 1.10E+10            | 1.10                                                   | 2.24   | 1.10          | 2.24   | 1.09         | 2.23   |
| Na-22   | 950.35              | 9.41                                                   | 18.72  | 9.37          | 18.64  | 9.39         | 18.69  |
| Al-26   | 2.66E+08            | 8.30                                                   | 16.45  | 5.61          | 10.06  | 8.27         | 16.38  |
| Si-32   | 237000              | 1.75                                                   | 3.00   | 1.44          | 2.23   | 2.04         | 3.74   |
| P-31    | 1.10E+10            | 2.26                                                   | 4.98   | 2.15          | 4.67   | 2.02         | 4.37   |
| P-32    | 14.28               | 46.71                                                  | 106.08 | 33.55         | 71.10  | 43.19        | 95.94  |
| P-33    | 25.3                | 12.88                                                  | 28.02  | 14.29         | 30.40  | 21.81        | 49.21  |
| S-35    | 87.4                | 24.04                                                  | 54.00  | 18.31         | 39.73  | 36.03        | 86.53  |
| Cl-36   | 1.10E+08            | 69.81                                                  | 161.69 | 56.89         | 126.12 | 88.30        | 212.21 |
| Ar-37   | 35                  | 12.70                                                  | 30.14  | 62.97         | 157.39 | 42.35        | 108.30 |
| Ar-39   | 98200               | 95.70                                                  | 222.23 | 96.72         | 227.13 | 245.05       | 610.57 |

Table 9.1: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in tellurium by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |               |       |              |       |
|---------|---------------------|--------------------------------------------------------|-------|---------------|-------|--------------|-------|
|         |                     | Semi-empirical formulae [9][10]                        |       | MENDL-2p [15] |       | MENDL-2 [16] |       |
|         |                     | [11]                                                   | [12]  | [11]          | [12]  | [11]         | [12]  |
| H-3     | 4503.4              | 31.02                                                  | 31.34 | 31.02         | 31.34 | 31.02        | 31.34 |
| Be-7    | 53.3                | 0.55                                                   | 0.21  | 0.55          | 0.21  | 0.55         | 0.21  |
| Be-10   | 5.84E+08            | 0.71                                                   | 9.66  | 0.71          | 9.66  | 0.71         | 9.66  |
| B-11    | 1.10E+10            | 0.00                                                   | 0.01  | 0.00          | 0.01  | 0.00         | 0.01  |
| C-14    | 2.09E+06            | 0.64                                                   | 0.17  | 0.64          | 0.17  | 0.64         | 0.17  |
| Ne-20   | 1.10E+10            | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Na-22   | 950.35              | 0.06                                                   | 0.02  | 0.06          | 0.02  | 0.06         | 0.02  |
| Al-26   | 2.66E+08            | 0.01                                                   | 0.01  | 0.01          | 0.01  | 0.01         | 0.01  |
| Si-32   | 237000              | 0.06                                                   | 0.02  | 0.06          | 0.02  | 0.06         | 0.02  |
| P-31    | 1.10E+10            | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| P-32    | 14.28               | 0.16                                                   | 0.04  | 0.16          | 0.04  | 0.16         | 0.04  |
| P-33    | 25.3                | 0.13                                                   | 0.04  | 0.13          | 0.04  | 0.13         | 0.04  |
| S-35    | 87.4                | 0.14                                                   | 0.04  | 0.14          | 0.04  | 0.14         | 0.04  |
| Cl-36   | 1.10E+08            | 0.07                                                   | 0.04  | 0.07          | 0.04  | 0.07         | 0.04  |
| Ar-37   | 35                  | 0.05                                                   | 0.01  | 0.05          | 0.01  | 0.05         | 0.01  |
| Ar-39   | 98200               | 0.19                                                   | 0.05  | 0.19          | 0.05  | 0.19         | 0.05  |
| Ar-42   | 12000               | 0.02                                                   | 0.01  | 0.02          | 0.01  | 0.02         | 0.01  |
| K-42    | 0.52                | 0.09                                                   | 0.02  | 0.09          | 0.02  | 0.09         | 0.02  |
| K-43    | 0.93                | 0.05                                                   | 0.01  | 0.05          | 0.01  | 0.05         | 0.01  |
| Ca-40   | 1.10E+10            | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ca-41   | 3.65E+07            | 0.02                                                   | 0.01  | 0.02          | 0.01  | 0.02         | 0.01  |
| Ca-45   | 165                 | 0.06                                                   | 0.02  | 0.06          | 0.02  | 0.06         | 0.02  |
| Ca-47   | 4.536               | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Sc-46   | 83.8                | 0.13                                                   | 0.04  | 0.13          | 0.04  | 0.13         | 0.04  |
| Sc-47   | 3.42                | 0.07                                                   | 0.02  | 0.07          | 0.02  | 0.07         | 0.02  |
| Sc-48   | 1.82                | 0.02                                                   | 0.01  | 0.02          | 0.01  | 0.02         | 0.01  |
| Ti-44   | 17200               | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| V-48    | 15.976              | 0.05                                                   | 0.01  | 0.05          | 0.01  | 0.05         | 0.01  |
| V-49    | 330                 | 0.16                                                   | 0.04  | 0.16          | 0.04  | 0.16         | 0.04  |
| V-51    | 1.10E+10            | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Cr-48   | 0.989               | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Cr-51   | 27.7                | 0.12                                                   | 0.03  | 0.12          | 0.03  | 0.12         | 0.03  |
| Mn-52   | 5.59                | 0.03                                                   | 0.01  | 0.03          | 0.01  | 0.03         | 0.01  |
| Mn-53   | 1.38E+09            | 0.01                                                   | 0.01  | 0.01          | 0.01  | 0.01         | 0.01  |
| Mn-54   | 312                 | 0.18                                                   | 0.05  | 0.18          | 0.05  | 0.18         | 0.05  |
| Fe-55   | 986.2               | 0.08                                                   | 0.02  | 0.08          | 0.02  | 0.08         | 0.02  |
| Fe-59   | 44.6                | 0.05                                                   | 0.02  | 0.05          | 0.02  | 0.05         | 0.02  |
| Fe-60   | 5.47E+08            | 0.00                                                   | 0.01  | 0.00          | 0.01  | 0.00         | 0.01  |
| Co-56   | 78.8                | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Co-57   | 271                 | 0.05                                                   | 0.02  | 0.05          | 0.02  | 0.05         | 0.02  |
| Co-58   | 70.8                | 0.09                                                   | 0.03  | 0.09          | 0.03  | 0.09         | 0.03  |
| Co-60   | 1925.2              | 0.06                                                   | 0.02  | 0.06          | 0.02  | 0.06         | 0.02  |
| Ni-56   | 6.1                 | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ni-57   | 1.5                 | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ni-59   | 2.92E+07            | 0.06                                                   | 0.03  | 0.06          | 0.03  | 0.06         | 0.03  |
| Ni-63   | 36524               | 0.07                                                   | 0.03  | 0.07          | 0.03  | 0.07         | 0.03  |
| Ni-66   | 2.28                | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Cu-63   | 1.10E+10            | 0.00                                                   | 0.01  | 0.00          | 0.01  | 0.00         | 0.01  |
| Cu-67   | 2.58                | 0.02                                                   | 0.01  | 0.02          | 0.01  | 0.02         | 0.01  |
| Zn-65   | 244.1               | 0.23                                                   | 0.09  | 0.23          | 0.09  | 0.23         | 0.09  |
| Zn-72   | 1.938               | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ga-67   | 3.263               | 0.22                                                   | 0.09  | 0.22          | 0.09  | 0.22         | 0.09  |
| Ga-69   | 1.10E+10            | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ga-71   | 0.59                | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Ge-68   | 270.8               | 0.09                                                   | 0.04  | 0.09          | 0.04  | 0.09         | 0.04  |
| Ge-69   | 1.625               | 0.16                                                   | 0.06  | 0.16          | 0.06  | 0.16         | 0.06  |
| Ge-71   | 11.2                | 0.32                                                   | 0.12  | 0.32          | 0.12  | 0.32         | 0.12  |
| As-72   | 1.083               | 0.21                                                   | 0.08  | 0.21          | 0.08  | 0.21         | 0.08  |
| As-73   | 80.3                | 0.34                                                   | 0.13  | 0.34          | 0.13  | 0.34         | 0.13  |
| As-74   | 17.8                | 0.11                                                   | 0.04  | 0.11          | 0.04  | 0.11         | 0.04  |

Table 9.2: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in tellurium by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |                    |               |      |              |      |
|---------|---------------------|--------------------------------------------------------|--------------------|---------------|------|--------------|------|
|         |                     | Semi-empirical formulae [9][10]                        |                    | MENDL-2p [15] |      | MENDL-2 [16] |      |
|         |                     | [11]                                                   | [12]               | [11]          | [12] | [11]         | [12] |
| As-76   | 1.096               | 0.02                                                   | 0.01               | 0.02          | 0.01 | 0.02         | 0.01 |
| As-77   | 1.62                | 0.01                                                   | 0.00               | 0.01          | 0.00 | 0.01         | 0.00 |
| Se-72   | 8.4                 | 0.05                                                   | 0.02               | 0.05          | 0.02 | 0.05         | 0.02 |
| Se-75   | 118.5               | 0.33                                                   | 0.13               | 0.33          | 0.13 | 0.33         | 0.13 |
| Se-79   | 2.38E+07            | 0.02                                                   | 0.01               | 0.02          | 0.01 | 0.02         | 0.01 |
| Se-80   | 1.10E+10            | 0.00                                                   | 0.00               | 0.00          | 0.00 | 0.00         | 0.00 |
| Br-76   | 0.67                | 0.19                                                   | 0.07               | 0.19          | 0.07 | 0.19         | 0.07 |
| Br-77   | 2.375               | 0.35                                                   | 0.13               | 0.35          | 0.13 | 0.35         | 0.13 |
| Br-82   | 1.47                | 0.01                                                   | 0.00               | 0.01          | 0.00 | 0.01         | 0.00 |
| Kr-79   | 1.46                | 0.32                                                   | 0.12               | 0.32          | 0.12 | 0.32         | 0.12 |
| Kr-81   | 7.66E+07            | 0.28                                                   | 0.19               | 0.28          | 0.19 | 0.28         | 0.19 |
| Kr-85   | 3908.1              | 0.01                                                   | 0.00               | 0.01          | 0.00 | 0.01         | 0.00 |
| Rb-83   | 86.2                | 0.22                                                   | 0.09               | 0.22          | 0.09 | 0.22         | 0.09 |
| Rb-84   | 32.9                | 0.09                                                   | 0.04               | 0.09          | 0.04 | 0.09         | 0.04 |
| Rb-86   | 18.8                | 0.02                                                   | 0.01               | 0.02          | 0.01 | 0.02         | 0.01 |
| Sr-82   | 25                  | 0.17                                                   | 0.07               | 0.17          | 0.07 | 0.17         | 0.07 |
| Sr-83   | 1.35                | 0.29                                                   | 0.13               | 0.29          | 0.13 | 0.29         | 0.13 |
| Sr-85   | 64.8                | 0.58                                                   | 0.26               | 0.58          | 0.26 | 0.58         | 0.26 |
| Sr-89   | 50.5                | 0.02                                                   | 0.01               | 0.02          | 0.01 | 0.02         | 0.01 |
| Sr-90   | 10600               | 0.01                                                   | 0.00               | 0.01          | 0.00 | 0.01         | 0.00 |
| Y-86    | 0.61                | 0.38                                                   | 0.18               | 0.38          | 0.18 | 0.38         | 0.18 |
| Y-87    | 3.35                | 0.64                                                   | 0.28               | 0.64          | 0.28 | 0.64         | 0.28 |
| Y-88    | 106.6               | 0.22                                                   | 0.10               | 0.22          | 0.10 | 0.22         | 0.10 |
| Y-89    | 1.10E+10            | 0.00                                                   | 0.00               | 0.00          | 0.00 | 0.00         | 0.00 |
| Y-90    | 2.67                | 0.05                                                   | 0.02               | 0.05          | 0.02 | 0.05         | 0.02 |
| Y-91    | 58.5                | 0.03                                                   | 0.01               | 0.03          | 0.01 | 0.03         | 0.01 |
| Zr-86   | 0.688               | 0.08                                                   | 0.03               | 0.08          | 0.03 | 0.08         | 0.03 |
| Zr-88   | 83.4                | 0.57                                                   | 0.26               | 0.57          | 0.26 | 0.57         | 0.26 |
| Zr-89   | 3.27                | 0.77                                                   | 0.39               | 0.77          | 0.39 | 0.77         | 0.39 |
| Zr-92   | 1.10E+10            | 0.00                                                   | 0.01               | 0.00          | 0.01 | 0.00         | 0.01 |
| Zr-93   | 5.48E+08            | 0.01                                                   | 0.03               | 0.01          | 0.03 | 0.01         | 0.03 |
| Zr-95   | 64                  | 0.01                                                   | 0.01               | 0.01          | 0.01 | 0.01         | 0.01 |
| Nb-91   | 3.65E+06            | 0.95                                                   | 0.53               | 0.95          | 0.53 | 0.95         | 0.53 |
| Nb-91   | 62                  | 0.63                                                   | 0.34               | 0.63          | 0.34 | 0.63         | 0.34 |
| Nb-93   | 4967                | 0.30                                                   | 0.17               | 0.30          | 0.17 | 0.30         | 0.17 |
| Nb-94   | 7.30E+06            | 0.12                                                   | 0.07               | 0.12          | 0.07 | 0.12         | 0.07 |
| Nb-95   | 35                  | 0.06                                                   | 0.04               | 0.06          | 0.04 | 0.06         | 0.04 |
| Nb-96   | 0.975               | 0.02                                                   | 0.01               | 0.02          | 0.01 | 0.02         | 0.01 |
| Mo-93   | 1.10E+06            | 0.77                                                   | 0.45               | 0.77          | 0.45 | 0.77         | 0.45 |
| Mo-99   | 2.75                | 0.01                                                   | 0.01               | 0.01          | 0.01 | 0.01         | 0.01 |
| Tc-95   | 61                  | 0.69                                                   | 0.45               | 0.69          | 0.45 | 0.69         | 0.45 |
| Tc-96   | 4.3                 | 0.74                                                   | 0.48               | 0.74          | 0.48 | 0.74         | 0.48 |
| Tc-97   | 90                  | 0.69                                                   | 0.45               | 0.69          | 0.45 | 0.69         | 0.45 |
| Tc-97   | 9.49E+08            | 0.05                                                   | 0.23               | 0.05          | 0.23 | 0.05         | 0.23 |
| Tc-98   | 1.53E+09            | 0.01                                                   | 0.08               | 0.01          | 0.08 | 0.01         | 0.08 |
| Tc-99   | 8.39E+07            | 0.10                                                   | 0.12               | 0.10          | 0.12 | 0.10         | 0.12 |
| Ru-97   | 2.88                | 0.69                                                   | 0.49               | 0.69          | 0.49 | 0.69         | 0.49 |
| Ru-102  | 1.10E+10            | 0.00                                                   | 0.01               | 0.00          | 0.01 | 0.00         | 0.01 |
| Ru-103  | 39.4                | 0.07                                                   | 0.06               | 0.07          | 0.06 | 0.07         | 0.06 |
| Ru-106  | 367                 | 0.00                                                   | 0.00               | 0.00          | 0.00 | 0.00         | 0.00 |
| Ru-99   | 15                  | 0.68                                                   | 0.52               | 0.68          | 0.52 | 0.68         | 0.52 |
| Rh-100  | 0.87                | 1.07                                                   | 0.83               | 1.07          | 0.82 | 1.07         | 0.83 |
| Rh-101  | 1205.3              | 1.84                                                   | 1.47               | 1.84          | 1.47 | 1.84         | 1.47 |
| Rh-102  | 206                 | 0.93                                                   | 0.74               | 0.93          | 0.74 | 0.93         | 0.74 |
| Rh-102  | 1059                | 0.93                                                   | 0.74               | 0.93          | 0.74 | 0.93         | 0.74 |
| Rh-105  | 1.48                | 0.13                                                   | 0.11               | 0.13          | 0.11 | 0.13         | 0.11 |
| Pd-100  | 3.6                 | 0.22                                                   | 0.18               | 0.22          | 0.18 | 0.22         | 0.18 |
| Pd-103  | 17                  | 2.00                                                   | 1.76               | 2.00          | 1.75 | 2.00         | 1.76 |
| Pd-107  | 2.38E+09            | 0.01                                                   | 0.06               | 0.01          | 0.06 | 0.01         | 0.06 |
| Pd-112  | 0.88                | 0.00                                                   | 0.00               | 0.00          | 0.00 | 0.00         | 0.00 |
| Ag-105  | 41.3                | 2.14                                                   | 2.08               | 2.15          | 2.09 | 2.14         | 2.08 |
| Ag-106  | 8.5                 | 1.87                                                   | 1.81 <sup>15</sup> | 1.87          | 1.80 | 1.87         | 1.81 |



Table 9.3: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in tellurium by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|-------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae [9][10]                        |       | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                   | [12]  | [11]          | [12]   | [11]         | [12]   |
| Ag-108  | 46400               | 0.78                                                   | 0.80  | 0.78          | 0.81   | 0.78         | 0.80   |
| Ag-109  | 1.10E+10            | 0.00                                                   | 0.03  | 0.00          | 0.03   | 0.00         | 0.03   |
| Ag-110  | 252                 | 0.19                                                   | 0.21  | 0.20          | 0.23   | 0.19         | 0.21   |
| Ag-111  | 7.45                | 0.10                                                   | 0.11  | 0.10          | 0.13   | 0.10         | 0.11   |
| Cd-109  | 453                 | 4.46                                                   | 5.14  | 4.55          | 5.39   | 4.46         | 5.14   |
| Cd-113  | 5113                | 0.18                                                   | 0.24  | 0.20          | 0.29   | 0.19         | 0.25   |
| Cd-115  | 44.8                | 0.04                                                   | 0.05  | 0.05          | 0.07   | 0.05         | 0.07   |
| In-111  | 2.83                | 3.23                                                   | 3.96  | 3.36          | 4.28   | 3.23         | 3.96   |
| In-114  | 49.5                | 0.75                                                   | 1.03  | 0.88          | 1.35   | 0.76         | 1.06   |
| Sn-113  | 115                 | 6.93                                                   | 10.06 | 7.46          | 11.45  | 6.96         | 10.14  |
| Sn-119  | 250                 | 1.47                                                   | 2.69  | 1.43          | 2.58   | 2.29         | 4.73   |
| Sn-120  | 1.10E+10            | 0.01                                                   | 0.10  | 0.01          | 0.12   | 0.02         | 0.36   |
| Sn-120  | 20100               | 1.99                                                   | 3.65  | 1.51          | 2.47   | 2.93         | 6.07   |
| Sn-123  | 129                 | 2.16                                                   | 3.94  | 1.52          | 2.35   | 2.97         | 6.09   |
| Sn-126  | 3.65E+07            | 0.34                                                   | 0.71  | 0.25          | 0.45   | 0.93         | 2.49   |
| Sb-119  | 1.58                | 12.08                                                  | 25.06 | 9.50          | 18.91  | 12.07        | 25.03  |
| Sb-120  | 5.76                | 14.67                                                  | 31.48 | 8.72          | 17.02  | 13.47        | 28.50  |
| Sb-122  | 2.68                | 20.24                                                  | 46.08 | 9.89          | 20.66  | 17.84        | 39.49  |
| Sb-124  | 60.2                | 23.77                                                  | 56.02 | 10.57         | 23.18  | 20.98        | 48.03  |
| Sb-125  | 986                 | 21.68                                                  | 51.84 | 11.11         | 25.43  | 24.30        | 56.96  |
| Sb-127  | 3.9                 | 13.23                                                  | 30.53 | 6.26          | 12.79  | 19.95        | 48.08  |
| Te-118  | 6                   | 15.92                                                  | 29.97 | 19.86         | 39.66  | 18.42        | 36.15  |
| Te-119  | 4.68                | 20.20                                                  | 39.56 | 23.84         | 48.44  | 22.98        | 46.49  |
| Te-121  | 154                 | 35.64                                                  | 77.18 | 45.50         | 101.83 | 45.98        | 103.65 |
| Te-123  | 119.7               | 11.68                                                  | 27.48 | 18.57         | 43.85  | 33.18        | 83.71  |
| Te-125  | 58                  | 22.89                                                  | 53.87 | 21.46         | 49.41  | 53.09        | 133.66 |
| Te-127  | 109                 | 33.53                                                  | 79.23 | 20.84         | 46.59  | 66.50        | 166.55 |
| Te-129  | 33.5                | 29.37                                                  | 68.94 | 15.09         | 31.83  | 58.14        | 142.96 |
| I-124   | 4.2                 | 28.98                                                  | 74.12 | 35.18         | 89.94  | 28.98        | 74.12  |
| I-125   | 60.2                | 36.48                                                  | 97.15 | 57.54         | 152.06 | 36.48        | 97.15  |
| I-126   | 13                  | 31.60                                                  | 87.48 | 35.37         | 98.05  | 31.60        | 87.48  |

Table 10: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in titanium by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae                                |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [9][10]                                                | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 41.11                                                  | 63.64  | 41.11         | 63.64  | 41.11        | 63.64  |
| Be-7    | 53.3                | 2.10                                                   | 1.76   | 2.10          | 1.76   | 2.10         | 1.76   |
| Be-10   | 5.84E+08            | 0.97                                                   | 0.80   | 0.97          | 0.80   | 0.97         | 0.80   |
| B-11    | 1.10E+10            | 0.05                                                   | 0.03   | 0.05          | 0.03   | 0.05         | 0.03   |
| C-14    | 2.09E+06            | 0.51                                                   | 0.28   | 0.51          | 0.28   | 0.51         | 0.28   |
| Ne-20   | 1.10E+10            | 0.08                                                   | 0.05   | 0.08          | 0.05   | 0.08         | 0.05   |
| Na-22   | 950.35              | 0.93                                                   | 0.66   | 0.93          | 0.66   | 0.93         | 0.66   |
| Al-26   | 2.66E+08            | 0.94                                                   | 0.81   | 0.94          | 0.81   | 0.94         | 0.81   |
| Si-32   | 237000              | 0.87                                                   | 1.05   | 0.87          | 1.05   | 0.86         | 1.02   |
| P-31    | 1.10E+10            | 0.27                                                   | 0.31   | 0.27          | 0.31   | 0.27         | 0.30   |
| P-32    | 14.28               | 5.37                                                   | 6.47   | 5.00          | 5.63   | 5.33         | 6.39   |
| P-33    | 25.3                | 3.14                                                   | 4.01   | 2.81          | 3.23   | 3.09         | 3.90   |
| S-35    | 87.4                | 5.79                                                   | 8.46   | 4.70          | 5.83   | 5.54         | 7.88   |
| Cl-36   | 1.10E+08            | 11.65                                                  | 18.33  | 9.80          | 13.94  | 11.17        | 17.22  |
| Ar-37   | 35                  | 7.22                                                   | 12.26  | 10.27         | 19.85  | 7.12         | 12.01  |
| Ar-39   | 98200               | 15.50                                                  | 28.89  | 10.04         | 15.69  | 14.19        | 25.80  |
| Ar-42   | 12000               | 0.81                                                   | 1.73   | 0.39          | 0.72   | 0.63         | 1.28   |
| K-42    | 0.52                | 13.04                                                  | 28.13  | 6.27          | 11.59  | 12.26        | 26.24  |
| K-43    | 0.93                | 5.44                                                   | 12.11  | 2.28          | 4.36   | 5.07         | 11.28  |
| Ca-40   | 1.10E+10            | 0.19                                                   | 0.38   | 1.05          | 2.48   | 0.39         | 0.88   |
| Ca-41   | 3.65E+07            | 14.17                                                  | 29.80  | 39.86         | 93.19  | 24.86        | 56.78  |
| Ca-45   | 165                 | 14.25                                                  | 33.90  | 5.58          | 12.27  | 23.92        | 58.00  |
| Ca-47   | 4.536               | 0.31                                                   | 0.66   | 0.23          | 0.46   | 3.49         | 8.98   |
| Sc-46   | 83.8                | 107.80                                                 | 259.32 | 49.97         | 113.36 | 115.95       | 291.08 |
| Sc-47   | 3.42                | 48.78                                                  | 111.34 | 32.03         | 68.25  | 96.70        | 239.45 |
| Sc-48   | 1.82                | 58.59                                                  | 177.24 | 54.20         | 166.04 | 25.49        | 69.29  |
| Ti-44   | 17200               | 4.84                                                   | 11.26  | 5.56          | 13.16  | 4.04         | 9.32   |
| V-48    | 15.976              | 54.33                                                  | 168.44 | 93.24         | 379.80 | 54.33        | 168.44 |
| V-49    | 330                 | 8.34                                                   | 24.23  | 12.30         | 49.90  | 8.34         | 24.23  |

Table 11.1: Calculations of production rates (in units of  $\text{kg}^{-1} \text{day}^{-1}$ ) of radioactive isotopes in xenon by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|-------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae [9][10]                       |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                  | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 31.141                                                | 30.727 | 31.141        | 30.727 | 31.141       | 30.727 |
| Be-7    | 53.3                | 0.548                                                 | 0.209  | 0.548         | 0.209  | 0.548        | 0.209  |
| Be-10   | 5.84E+08            | 4.590                                                 | 10.031 | 4.590         | 10.031 | 4.590        | 10.031 |
| B-11    | 1.10E+10            | 0.035                                                 | 0.009  | 0.035         | 0.009  | 0.035        | 0.009  |
| C-14    | 2.09E+06            | 0.632                                                 | 0.164  | 0.632         | 0.164  | 0.632        | 0.164  |
| Ne-20   | 1.10E+10            | 0.017                                                 | 0.003  | 0.017         | 0.003  | 0.017        | 0.003  |
| Na-22   | 950.35              | 0.069                                                 | 0.018  | 0.069         | 0.018  | 0.069        | 0.018  |
| Al-26   | 2.66E+08            | 0.044                                                 | 0.012  | 0.044         | 0.012  | 0.044        | 0.012  |
| Si-32   | 237000              | 0.050                                                 | 0.016  | 0.050         | 0.016  | 0.050        | 0.016  |
| P-31    | 1.10E+10            | 0.007                                                 | 0.002  | 0.007         | 0.002  | 0.007        | 0.002  |
| P-32    | 14.28               | 0.151                                                 | 0.041  | 0.151         | 0.041  | 0.151        | 0.041  |
| P-33    | 25.3                | 0.117                                                 | 0.034  | 0.117         | 0.034  | 0.117        | 0.034  |
| S-35    | 87.4                | 0.123                                                 | 0.035  | 0.123         | 0.035  | 0.123        | 0.035  |
| Cl-36   | 1.10E+08            | 0.149                                                 | 0.040  | 0.149         | 0.040  | 0.149        | 0.040  |
| Ar-37   | 35                  | 0.059                                                 | 0.016  | 0.059         | 0.016  | 0.059        | 0.016  |
| Ar-39   | 98200               | 0.163                                                 | 0.046  | 0.163         | 0.046  | 0.163        | 0.046  |
| Ar-42   | 12000               | 0.014                                                 | 0.005  | 0.014         | 0.005  | 0.014        | 0.005  |
| K-42    | 0.52                | 0.074                                                 | 0.021  | 0.074         | 0.021  | 0.074        | 0.021  |
| K-43    | 0.93                | 0.040                                                 | 0.013  | 0.040         | 0.013  | 0.040        | 0.013  |
| Ca-40   | 1.10E+10            | 0.001                                                 | 0.000  | 0.001         | 0.000  | 0.001        | 0.000  |
| Ca-41   | 3.65E+07            | 0.036                                                 | 0.010  | 0.036         | 0.010  | 0.036        | 0.010  |
| Ca-45   | 165                 | 0.046                                                 | 0.015  | 0.046         | 0.015  | 0.046        | 0.015  |
| Ca-47   | 4.536               | 0.007                                                 | 0.003  | 0.007         | 0.003  | 0.007        | 0.003  |
| Sc-46   | 83.8                | 0.106                                                 | 0.030  | 0.106         | 0.030  | 0.106        | 0.030  |
| Sc-47   | 3.42                | 0.057                                                 | 0.017  | 0.057         | 0.017  | 0.057        | 0.017  |
| Sc-48   | 1.82                | 0.020                                                 | 0.007  | 0.020         | 0.007  | 0.020        | 0.007  |
| Ti-44   | 17200               | 0.005                                                 | 0.001  | 0.005         | 0.001  | 0.005        | 0.001  |
| V-48    | 15.976              | 0.063                                                 | 0.017  | 0.063         | 0.017  | 0.063        | 0.017  |
| V-49    | 330                 | 0.190                                                 | 0.052  | 0.190         | 0.052  | 0.190        | 0.052  |
| V-51    | 1.10E+10            | 0.005                                                 | 0.002  | 0.005         | 0.002  | 0.005        | 0.002  |
| Cr-48   | 0.989               | 0.002                                                 | 0.001  | 0.002         | 0.001  | 0.002        | 0.001  |
| Cr-51   | 27.7                | 0.150                                                 | 0.041  | 0.150         | 0.041  | 0.150        | 0.041  |
| Mn-52   | 5.59                | 0.039                                                 | 0.011  | 0.039         | 0.011  | 0.039        | 0.011  |
| Mn-53   | 1.38E+09            | 0.055                                                 | 0.015  | 0.055         | 0.015  | 0.055        | 0.015  |
| Mn-54   | 312                 | 0.176                                                 | 0.049  | 0.176         | 0.049  | 0.176        | 0.049  |
| Fe-55   | 986.2               | 0.104                                                 | 0.029  | 0.104         | 0.029  | 0.104        | 0.029  |
| Fe-59   | 44.6                | 0.036                                                 | 0.013  | 0.036         | 0.013  | 0.036        | 0.013  |
| Fe-60   | 5.47E+08            | 0.015                                                 | 0.005  | 0.015         | 0.005  | 0.015        | 0.005  |
| Co-56   | 78.8                | 0.022                                                 | 0.006  | 0.022         | 0.006  | 0.022        | 0.006  |
| Co-57   | 271                 | 0.114                                                 | 0.045  | 0.114         | 0.045  | 0.114        | 0.045  |
| Co-58   | 70.8                | 0.198                                                 | 0.077  | 0.198         | 0.077  | 0.198        | 0.077  |
| Co-60   | 1925.2              | 0.084                                                 | 0.032  | 0.084         | 0.032  | 0.084        | 0.032  |
| Ni-56   | 6.1                 | 0.000                                                 | 0.000  | 0.000         | 0.000  | 0.000        | 0.000  |
| Ni-57   | 1.5                 | 0.004                                                 | 0.002  | 0.004         | 0.002  | 0.004        | 0.002  |
| Ni-59   | 2.92E+07            | 0.082                                                 | 0.032  | 0.082         | 0.032  | 0.082        | 0.032  |
| Ni-63   | 36524               | 0.058                                                 | 0.021  | 0.058         | 0.021  | 0.058        | 0.021  |
| Ni-66   | 2.28                | 0.006                                                 | 0.002  | 0.006         | 0.002  | 0.006        | 0.002  |
| Cu-63   | 1.10E+10            | 0.015                                                 | 0.006  | 0.015         | 0.006  | 0.015        | 0.006  |
| Cu-67   | 2.58                | 0.016                                                 | 0.006  | 0.016         | 0.006  | 0.016        | 0.006  |
| Zn-65   | 244.1               | 0.264                                                 | 0.102  | 0.264         | 0.102  | 0.264        | 0.102  |
| Zn-72   | 1.938               | 0.002                                                 | 0.001  | 0.002         | 0.001  | 0.002        | 0.001  |
| Ga-67   | 3.263               | 0.259                                                 | 0.100  | 0.259         | 0.100  | 0.259        | 0.100  |
| Ga-69   | 1.10E+10            | 0.007                                                 | 0.003  | 0.007         | 0.003  | 0.007        | 0.003  |
| Ga-71   | 0.59                | 0.009                                                 | 0.003  | 0.009         | 0.003  | 0.009        | 0.003  |
| Ge-68   | 270.8               | 0.116                                                 | 0.046  | 0.116         | 0.046  | 0.116        | 0.046  |
| Ge-69   | 1.625               | 0.204                                                 | 0.079  | 0.204         | 0.079  | 0.204        | 0.079  |
| Ge-71   | 11.2                | 0.312                                                 | 0.121  | 0.312         | 0.121  | 0.312        | 0.121  |
| As-72   | 1.083               | 0.213                                                 | 0.082  | 0.213         | 0.082  | 0.213        | 0.082  |
| As-73   | 80.3                | 0.311                                                 | 0.120  | 0.311         | 0.120  | 0.311        | 0.120  |
| As-74   | 17.8                | 0.072                                                 | 0.027  | 0.072         | 0.027  | 0.072        | 0.027  |

Table 11.2: Calculations of production rates (in units of  $\text{kg}^{-1} \text{day}^{-1}$ ) of radioactive isotopes in xenon by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{day}^{-1}$ ) |       |               |       |              |       |
|---------|---------------------|-------------------------------------------------------|-------|---------------|-------|--------------|-------|
|         |                     | Semi-empirical formulae [9][10]                       |       | MENDL-2p [15] |       | MENDL-2 [16] |       |
|         |                     | [11]                                                  | [12]  | [11]          | [12]  | [11]         | [12]  |
| As-76   | 1.096               | 0.016                                                 | 0.006 | 0.016         | 0.006 | 0.016        | 0.006 |
| As-77   | 1.62                | 0.010                                                 | 0.004 | 0.010         | 0.004 | 0.010        | 0.004 |
| Se-72   | 8.4                 | 0.065                                                 | 0.026 | 0.065         | 0.026 | 0.065        | 0.026 |
| Se-75   | 118.5               | 0.308                                                 | 0.118 | 0.308         | 0.118 | 0.308        | 0.118 |
| Se-79   | 2.38E+07            | 0.013                                                 | 0.005 | 0.013         | 0.005 | 0.013        | 0.005 |
| Se-80   | 1.10E+10            | 0.000                                                 | 0.000 | 0.000         | 0.000 | 0.000        | 0.000 |
| Br-76   | 0.67                | 0.230                                                 | 0.087 | 0.230         | 0.087 | 0.230        | 0.087 |
| Br-77   | 2.375               | 0.329                                                 | 0.124 | 0.329         | 0.124 | 0.329        | 0.124 |
| Br-82   | 1.47                | 0.007                                                 | 0.002 | 0.007         | 0.002 | 0.007        | 0.002 |
| Kr-79   | 1.46                | 0.316                                                 | 0.118 | 0.316         | 0.118 | 0.316        | 0.118 |
| Kr-81   | 7.66E+07            | 0.417                                                 | 0.155 | 0.417         | 0.155 | 0.417        | 0.155 |
| Kr-85   | 3908.1              | 0.006                                                 | 0.002 | 0.006         | 0.002 | 0.006        | 0.002 |
| Rb-83   | 86.2                | 0.135                                                 | 0.050 | 0.135         | 0.050 | 0.135        | 0.050 |
| Rb-84   | 32.9                | 0.055                                                 | 0.020 | 0.055         | 0.020 | 0.055        | 0.020 |
| Rb-86   | 18.8                | 0.014                                                 | 0.005 | 0.014         | 0.005 | 0.014        | 0.005 |
| Sr-82   | 25                  | 0.186                                                 | 0.072 | 0.186         | 0.072 | 0.186        | 0.072 |
| Sr-83   | 1.35                | 0.294                                                 | 0.114 | 0.294         | 0.114 | 0.294        | 0.114 |
| Sr-85   | 64.8                | 0.455                                                 | 0.182 | 0.455         | 0.182 | 0.455        | 0.182 |
| Sr-89   | 50.5                | 0.013                                                 | 0.005 | 0.013         | 0.005 | 0.013        | 0.005 |
| Sr-90   | 10600               | 0.008                                                 | 0.003 | 0.008         | 0.003 | 0.008        | 0.003 |
| Y-86    | 0.61                | 0.309                                                 | 0.127 | 0.309         | 0.127 | 0.309        | 0.127 |
| Y-87    | 3.35                | 0.488                                                 | 0.190 | 0.488         | 0.190 | 0.488        | 0.190 |
| Y-88    | 106.6               | 0.123                                                 | 0.051 | 0.123         | 0.051 | 0.123        | 0.051 |
| Y-89    | 1.10E+10            | 0.004                                                 | 0.002 | 0.004         | 0.002 | 0.004        | 0.002 |
| Y-90    | 2.67                | 0.032                                                 | 0.012 | 0.032         | 0.012 | 0.032        | 0.012 |
| Y-91    | 58.5                | 0.022                                                 | 0.008 | 0.022         | 0.008 | 0.022        | 0.008 |
| Zr-86   | 0.688               | 0.101                                                 | 0.033 | 0.101         | 0.033 | 0.101        | 0.033 |
| Zr-88   | 83.4                | 0.582                                                 | 0.239 | 0.582         | 0.239 | 0.582        | 0.239 |
| Zr-89   | 3.27                | 0.687                                                 | 0.314 | 0.687         | 0.314 | 0.687        | 0.314 |
| Zr-92   | 1.10E+10            | 0.006                                                 | 0.003 | 0.006         | 0.003 | 0.006        | 0.003 |
| Zr-93   | 5.48E+08            | 0.025                                                 | 0.012 | 0.025         | 0.012 | 0.025        | 0.012 |
| Zr-95   | 64                  | 0.006                                                 | 0.003 | 0.006         | 0.003 | 0.006        | 0.003 |
| Nb-91   | 3.65E+06            | 0.813                                                 | 0.396 | 0.813         | 0.396 | 0.813        | 0.396 |
| Nb-91   | 62                  | 0.481                                                 | 0.229 | 0.481         | 0.229 | 0.481        | 0.229 |
| Nb-93   | 4967                | 0.162                                                 | 0.080 | 0.162         | 0.080 | 0.162        | 0.080 |
| Nb-94   | 7.30E+06            | 0.064                                                 | 0.032 | 0.064         | 0.032 | 0.064        | 0.032 |
| Nb-95   | 35                  | 0.033                                                 | 0.017 | 0.033         | 0.017 | 0.033        | 0.017 |
| Nb-96   | 0.975               | 0.011                                                 | 0.006 | 0.011         | 0.006 | 0.011        | 0.006 |
| Mo-93   | 1.10E+06            | 0.612                                                 | 0.308 | 0.612         | 0.308 | 0.612        | 0.308 |
| Mo-99   | 2.75                | 0.004                                                 | 0.002 | 0.004         | 0.002 | 0.004        | 0.002 |
| Tc-95   | 61                  | 0.644                                                 | 0.365 | 0.644         | 0.365 | 0.644        | 0.365 |
| Tc-96   | 4.3                 | 0.510                                                 | 0.293 | 0.510         | 0.293 | 0.510        | 0.293 |
| Tc-97   | 90                  | 0.376                                                 | 0.220 | 0.376         | 0.220 | 0.376        | 0.220 |
| Tc-97   | 9.49E+08            | 0.195                                                 | 0.114 | 0.195         | 0.114 | 0.195        | 0.114 |
| Tc-98   | 1.53E+09            | 0.058                                                 | 0.035 | 0.058         | 0.035 | 0.058        | 0.035 |
| Tc-99   | 8.39E+07            | 0.087                                                 | 0.054 | 0.087         | 0.054 | 0.087        | 0.054 |
| Ru-97   | 2.88                | 0.713                                                 | 0.437 | 0.713         | 0.437 | 0.713        | 0.437 |
| Ru-102  | 1.10E+10            | 0.004                                                 | 0.003 | 0.004         | 0.003 | 0.004        | 0.003 |
| Ru-103  | 39.4                | 0.031                                                 | 0.022 | 0.031         | 0.022 | 0.031        | 0.022 |
| Ru-106  | 367                 | 0.002                                                 | 0.001 | 0.002         | 0.001 | 0.002        | 0.001 |
| Rh-99   | 15                  | 0.715                                                 | 0.471 | 0.715         | 0.471 | 0.715        | 0.471 |
| Rh-100  | 0.87                | 0.925                                                 | 0.620 | 0.925         | 0.620 | 0.925        | 0.620 |
| Rh-101  | 1205.3              | 1.430                                                 | 0.993 | 1.430         | 0.993 | 1.430        | 0.993 |
| Rh-102  | 206                 | 0.483                                                 | 0.335 | 0.483         | 0.335 | 0.483        | 0.335 |
| Rh-102  | 1059                | 0.483                                                 | 0.335 | 0.483         | 0.335 | 0.483        | 0.335 |
| Rh-105  | 1.48                | 0.057                                                 | 0.043 | 0.057         | 0.043 | 0.057        | 0.043 |
| Pd-100  | 3.6                 | 0.256                                                 | 0.177 | 0.256         | 0.177 | 0.256        | 0.177 |
| Pd-103  | 17                  | 1.612                                                 | 1.210 | 1.611         | 1.210 | 1.612        | 1.210 |
| Pd-107  | 2.38E+09            | 0.026                                                 | 0.021 | 0.026         | 0.021 | 0.026        | 0.021 |
| Pd-112  | 0.88                | 0.001                                                 | 0.001 | 0.001         | 0.001 | 0.001        | 0.001 |
| Ag-105  | 41.3                | 1.802                                                 | 1.475 | 1.801         | 1.472 | 1.802        | 1.475 |
| Ag-106  | 8.5                 | 1.084                                                 | 0.893 | 1.083         | 0.891 | 1.084        | 0.893 |

Table 11.3: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in xenon by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |         |              |         |
|---------|---------------------|--------------------------------------------------------|--------|---------------|---------|--------------|---------|
|         |                     | Semi-empirical formulae [9][10]                        |        | MENDL-2p [15] |         | MENDL-2 [16] |         |
|         |                     | [11]                                                   | [12]   | [11]          | [12]    | [11]         | [12]    |
| Ag-108  | 46400               | 0.344                                                  | 0.300  | 0.344         | 0.299   | 0.344        | 0.300   |
| Ag-109  | 1.10E+10            | 0.010                                                  | 0.009  | 0.010         | 0.009   | 0.010        | 0.009   |
| Ag-110  | 252                 | 0.077                                                  | 0.073  | 0.077         | 0.072   | 0.077        | 0.073   |
| Ag-111  | 7.45                | 0.037                                                  | 0.037  | 0.037         | 0.037   | 0.037        | 0.037   |
| Cd-109  | 453                 | 3.084                                                  | 2.983  | 3.079         | 2.970   | 3.084        | 2.983   |
| Cd-113  | 5113                | 0.068                                                  | 0.073  | 0.068         | 0.073   | 0.068        | 0.073   |
| Cd-115  | 44.8                | 0.013                                                  | 0.015  | 0.013         | 0.015   | 0.013        | 0.015   |
| In-111  | 2.83                | 1.651                                                  | 1.691  | 1.655         | 1.703   | 1.651        | 1.691   |
| In-114  | 49.5                | 0.279                                                  | 0.318  | 0.283         | 0.329   | 0.279        | 0.318   |
| Sn-113  | 115                 | 4.387                                                  | 5.142  | 4.494         | 5.427   | 4.387        | 5.143   |
| Sn-119  | 250                 | 0.064                                                  | 0.094  | 0.070         | 0.107   | 0.075        | 0.119   |
| Sn-120  | 1.10E+10            | 0.002                                                  | 0.002  | 0.002         | 0.003   | 0.003        | 0.005   |
| Sn-120  | 20100               | 0.012                                                  | 0.018  | 0.013         | 0.020   | 0.018        | 0.032   |
| Sn-123  | 129                 | 0.004                                                  | 0.002  | 0.004         | 0.003   | 0.005        | 0.006   |
| Sn-126  | 3.65E+07            | 0.001                                                  | 0.000  | 0.001         | 0.000   | 0.001        | 0.001   |
| Sb-119  | 1.58                | 0.584                                                  | 0.867  | 0.776         | 1.346   | 0.617        | 0.940   |
| Sb-120  | 5.76                | 0.303                                                  | 0.470  | 0.389         | 0.684   | 0.356        | 0.589   |
| Sb-122  | 2.68                | 0.071                                                  | 0.101  | 0.109         | 0.197   | 0.139        | 0.260   |
| Sb-124  | 60.2                | 0.030                                                  | 0.014  | 0.042         | 0.044   | 0.066        | 0.100   |
| Sb-125  | 986                 | 0.016                                                  | 0.007  | 0.025         | 0.028   | 0.046        | 0.079   |
| Sb-127  | 3.9                 | 0.009                                                  | 0.002  | 0.010         | 0.006   | 0.016        | 0.020   |
| Te-118  | 6                   | 8.705                                                  | 13.541 | 9.958         | 16.797  | 8.733        | 13.607  |
| Te-119  | 4.68                | 11.063                                                 | 18.202 | 12.445        | 21.714  | 11.146       | 18.388  |
| Te-121  | 154                 | 25.876                                                 | 47.755 | 26.069        | 48.336  | 25.724       | 47.446  |
| Te-123  | 119.7               | 1.273                                                  | 2.332  | 1.307         | 2.403   | 2.159        | 4.563   |
| Te-125  | 58                  | 2.138                                                  | 3.927  | 1.608         | 2.620   | 3.111        | 6.477   |
| Te-127  | 109                 | 1.098                                                  | 1.992  | 0.806         | 1.271   | 1.916        | 4.127   |
| Te-129  | 33.5                | 0.615                                                  | 1.108  | 0.449         | 0.698   | 1.143        | 2.466   |
| Te-131  | 1.25                | 0.173                                                  | 0.306  | 0.132         | 0.204   | 0.655        | 1.563   |
| Te-132  | 3.25                | 0.103                                                  | 0.179  | 0.080         | 0.122   | 0.304        | 0.698   |
| I-124   | 4.2                 | 14.746                                                 | 31.783 | 9.030         | 18.095  | 13.537       | 28.730  |
| I-125   | 60.2                | 37.354                                                 | 80.640 | 37.806        | 82.503  | 39.183       | 84.885  |
| I-126   | 13                  | 21.343                                                 | 48.343 | 9.509         | 19.123  | 19.734       | 44.203  |
| I-131   | 8.04                | 14.644                                                 | 39.418 | 11.453        | 31.321  | 13.529       | 32.682  |
| I-133   | 0.87                | 3.416                                                  | 7.874  | 1.680         | 3.443   | 5.321        | 12.866  |
| Xe-122  | 0.84                | 7.983                                                  | 14.131 | 14.801        | 30.793  | 9.369        | 17.320  |
| Xe-125  | 0.71                | 3.703                                                  | 7.569  | 14.265        | 33.211  | 14.147       | 34.165  |
| Xe-127  | 36.4                | 35.722                                                 | 89.586 | 66.461        | 168.372 | 62.299       | 160.450 |
| Xe-129  | 8.89                | 20.550                                                 | 48.860 | 17.934        | 42.112  | 43.492       | 111.247 |
| Xe-131  | 11.77               | 21.691                                                 | 51.236 | 14.064        | 31.316  | 44.530       | 110.402 |
| Xe-133  | 5.25                | 13.757                                                 | 32.299 | 9.688         | 21.830  | 24.681       | 61.155  |
| Cs-131  | 9.69                | 32.756                                                 | 90.541 | 42.346        | 114.790 | 32.756       | 90.541  |
| Cs-132  | 6.47                | 16.774                                                 | 49.279 | 15.072        | 44.861  | 16.774       | 49.279  |
| Cs-133  | 1.10E+10            | 0.809                                                  | 2.159  | 0.934         | 2.474   | 0.809        | 2.159   |
| Cs-134  | 752                 | 10.161                                                 | 29.026 | 6.827         | 20.080  | 10.161       | 29.026  |
| Cs-135  | 8.40E+08            | 7.891                                                  | 19.668 | 6.242         | 15.300  | 11.637       | 29.496  |
| Cs-136  | 13.16               | 2.989                                                  | 9.344  | 1.099         | 4.257   | 2.989        | 9.344   |

Table 12: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in carbon by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae                                |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 0.00                                                   | 0.00   | 0.00          | 0.00   | 0.00         | 0.00   |
| Be-7    | 53.3                | 68.66                                                  | 150.32 | 68.66         | 150.32 | 68.66        | 150.32 |
| Be-10   | 5.84E+08            | 4.37                                                   | 7.76   | 4.37          | 7.76   | 4.37         | 7.76   |
| B-11    | 5.84E+10            | 6.74                                                   | 14.52  | 6.74          | 14.52  | 6.74         | 14.52  |

Table 13: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$ ) of radioactive isotopes in chromium by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |         |               |         |              |         |
|---------|---------------------|--------------------------------------------------------|---------|---------------|---------|--------------|---------|
|         |                     | Semi-empirical formulae [9][10]                        |         | MENDL-2p [15] |         | MENDL-2 [16] |         |
|         |                     | [11]                                                   | [12]    | [11]          | [12]    | [11]         | [12]    |
| H-3     | 4503.4              | 39.354                                                 | 59.432  | 39.354        | 59.432  | 39.354       | 59.432  |
| Be-7    | 53.3                | 2.002                                                  | 1.658   | 2.002         | 1.658   | 2.002        | 1.658   |
| Be-10   | 5.84E+08            | 0.803                                                  | 0.656   | 0.803         | 0.656   | 0.803        | 0.656   |
| B-11    | 1.10E+10            | 0.049                                                  | 0.026   | 0.049         | 0.026   | 0.049        | 0.026   |
| C-14    | 2.09E+06            | 0.445                                                  | 0.233   | 0.445         | 0.233   | 0.445        | 0.233   |
| Ne-20   | 1.10E+10            | 0.054                                                  | 0.030   | 0.054         | 0.030   | 0.054        | 0.030   |
| Na-22   | 950.35              | 0.643                                                  | 0.387   | 0.643         | 0.387   | 0.643        | 0.387   |
| Al-26   | 2.66E+08            | 0.606                                                  | 0.426   | 0.606         | 0.426   | 0.606        | 0.426   |
| Si-32   | 237000              | 0.483                                                  | 0.452   | 0.483         | 0.452   | 0.483        | 0.452   |
| P-31    | 1.10E+10            | 0.157                                                  | 0.141   | 0.157         | 0.141   | 0.157        | 0.141   |
| P-32    | 14.28               | 3.054                                                  | 2.866   | 3.054         | 2.866   | 3.054        | 2.866   |
| P-33    | 25.3                | 1.688                                                  | 1.675   | 1.688         | 1.675   | 1.688        | 1.675   |
| S-35    | 87.4                | 3.551                                                  | 3.993   | 3.551         | 3.993   | 3.523        | 3.933   |
| Cl-36   | 1.10E+08            | 5.661                                                  | 6.821   | 5.661         | 5.501   | 5.652        | 6.802   |
| Ar-37   | 35                  | 3.977                                                  | 5.155   | 3.977         | 5.868   | 3.912        | 5.009   |
| Ar-39   | 98200               | 7.437                                                  | 10.893  | 7.437         | 7.357   | 7.114        | 10.145  |
| Ar-42   | 12000               | 0.316                                                  | 0.556   | 0.316         | 0.422   | 0.291        | 0.498   |
| K-42    | 0.52                | 3.855                                                  | 6.786   | 3.855         | 3.899   | 3.628        | 6.244   |
| K-43    | 0.93                | 1.556                                                  | 2.894   | 1.556         | 1.536   | 1.426        | 2.584   |
| Ca-40   | 1.10E+10            | 0.079                                                  | 0.126   | 0.079         | 0.658   | 0.087        | 0.144   |
| Ca-41   | 3.65E+07            | 5.521                                                  | 9.287   | 5.521         | 24.249  | 6.009        | 10.398  |
| Ca-45   | 165                 | 4.107                                                  | 8.436   | 4.107         | 4.045   | 3.592        | 7.144   |
| Ca-47   | 4.536               | 0.459                                                  | 1.021   | 0.459         | 0.569   | 0.330        | 0.702   |
| Sc-46   | 83.8                | 19.214                                                 | 41.445  | 19.214        | 21.979  | 19.176       | 41.343  |
| Sc-47   | 3.42                | 8.792                                                  | 19.633  | 8.792         | 7.128   | 8.792        | 19.848  |
| Sc-48   | 1.82                | 1.503                                                  | 3.474   | 1.503         | 1.807   | 1.960        | 4.666   |
| Ti-44   | 17200               | 1.570                                                  | 3.140   | 1.570         | 5.039   | 1.518        | 3.007   |
| V-48    | 15.976              | 59.465                                                 | 138.360 | 59.465        | 147.652 | 62.855       | 145.746 |
| V-49    | 330                 | 109.580                                                | 256.800 | 109.580       | 261.137 | 134.158      | 321.027 |
| V-51    | 1.10E+10            | 4.062                                                  | 8.915   | 4.062         | 5.221   | 8.006        | 19.411  |
| Cr-48   | 0.989               | 2.891                                                  | 6.718   | 2.891         | 7.291   | 2.223        | 5.044   |
| Cr-51   | 27.7                | 122.963                                                | 295.247 | 122.963       | 501.106 | 187.724      | 465.852 |
| Mn-52   | 5.59                | 54.479                                                 | 168.458 | 54.479        | 346.881 | 54.479       | 168.458 |
| Mn-53   | 1.38E+09            | 3.275                                                  | 9.953   | 3.275         | 21.647  | 3.275        | 9.953   |
| Mn-54   | 312                 | 1.823                                                  | 5.699   | 1.823         | 12.372  | 1.823        | 5.699   |

Table 14: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in iron by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae [9][10]                        |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 37.06                                                  | 55.97  | 37.06         | 55.97  | 37.06        | 55.97  |
| Be-7    | 53.3                | 1.95                                                   | 1.63   | 1.95          | 1.63   | 1.95         | 1.63   |
| Be-10   | 5.84E+08            | 0.62                                                   | 0.52   | 0.62          | 0.52   | 0.62         | 0.52   |
| B-11    | 1.10E+10            | 0.05                                                   | 0.02   | 0.05          | 0.02   | 0.05         | 0.02   |
| C-14    | 2.09E+06            | 0.44                                                   | 0.21   | 0.44          | 0.21   | 0.44         | 0.21   |
| Ne-20   | 1.10E+10            | 0.04                                                   | 0.02   | 0.04          | 0.02   | 0.04         | 0.02   |
| Na-22   | 950.35              | 0.39                                                   | 0.25   | 0.39          | 0.25   | 0.39         | 0.25   |
| Al-26   | 2.66E+08            | 0.34                                                   | 0.26   | 0.34          | 0.26   | 0.34         | 0.26   |
| Si-32   | 237000              | 0.38                                                   | 0.22   | 0.38          | 0.22   | 0.38         | 0.22   |
| P-31    | 1.10E+10            | 0.08                                                   | 0.07   | 0.08          | 0.07   | 0.08         | 0.07   |
| P-32    | 14.28               | 2.05                                                   | 1.45   | 2.05          | 1.45   | 2.05         | 1.45   |
| P-33    | 25.3                | 1.36                                                   | 0.85   | 1.36          | 0.85   | 1.36         | 0.85   |
| S-35    | 87.4                | 2.80                                                   | 1.94   | 2.80          | 1.94   | 2.80         | 1.94   |
| Cl-36   | 1.10E+08            | 3.13                                                   | 3.22   | 3.13          | 3.22   | 3.13         | 3.22   |
| Ar-37   | 35                  | 1.70                                                   | 2.39   | 1.70          | 2.39   | 1.70         | 2.39   |
| Ar-39   | 98200               | 5.02                                                   | 4.49   | 5.02          | 4.49   | 5.02         | 4.49   |
| Ar-42   | 12000               | 0.22                                                   | 0.19   | 0.22          | 0.19   | 0.22         | 0.19   |
| K-42    | 0.52                | 2.41                                                   | 2.33   | 2.03          | 1.70   | 2.41         | 2.33   |
| K-43    | 0.93                | 0.94                                                   | 0.94   | 0.77          | 0.66   | 0.94         | 0.94   |
| Ca-40   | 1.10E+10            | 0.02                                                   | 0.05   | 0.07          | 0.17   | 0.02         | 0.05   |
| Ca-41   | 3.65E+07            | 1.81                                                   | 3.58   | 3.07          | 6.51   | 1.81         | 3.58   |
| Ca-45   | 165                 | 2.18                                                   | 2.44   | 1.61          | 1.55   | 2.18         | 2.44   |
| Ca-47   | 4.536               | 0.23                                                   | 0.26   | 0.18          | 0.19   | 0.23         | 0.26   |
| Sc-46   | 83.8                | 8.58                                                   | 10.99  | 5.93          | 6.88   | 8.58         | 10.97  |
| Sc-47   | 3.42                | 4.02                                                   | 4.92   | 2.42          | 2.60   | 4.02         | 4.92   |
| Sc-48   | 1.82                | 0.62                                                   | 0.78   | 0.42          | 0.52   | 0.62         | 0.78   |
| Ti-44   | 17200               | 0.31                                                   | 1.01   | 0.78          | 2.00   | 0.31         | 1.01   |
| V-48    | 15.976              | 11.28                                                  | 33.37  | 18.94         | 46.31  | 11.28        | 33.35  |
| V-49    | 330                 | 42.65                                                  | 106.34 | 39.74         | 88.34  | 42.65        | 106.29 |
| V-51    | 1.10E+10            | 1.44                                                   | 2.10   | 0.57          | 0.84   | 1.44         | 2.10   |
| Cr-48   | 0.989               | 0.40                                                   | 1.79   | 1.28          | 3.35   | 0.40         | 1.79   |
| Cr-51   | 27.7                | 73.60                                                  | 205.35 | 63.34         | 154.15 | 74.47        | 204.78 |
| Mn-52   | 5.59                | 30.97                                                  | 103.45 | 50.24         | 127.48 | 40.10        | 111.77 |
| Mn-53   | 1.38E+09            | 42.19                                                  | 95.80  | 46.26         | 110.90 | 50.49        | 120.62 |
| Mn-54   | 312                 | 95.13                                                  | 245.74 | 69.52         | 174.18 | 130.28       | 339.38 |
| Fe-55   | 986.2               | 139.01                                                 | 297.74 | 206.28        | 515.48 | 232.04       | 567.90 |
| Co-56   | 78.8                | 47.76                                                  | 151.15 | 73.58         | 309.01 | 47.76        | 151.15 |
| Co-57   | 271                 | 0.00                                                   | 4.47   | 0.00          | 9.48   | 0.00         | 4.47   |
| Co-58   | 70.8                | 0.00                                                   | 0.61   | 0.00          | 1.27   | 0.00         | 0.61   |

Table 15: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in manganese by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Semi-empirical formulae [9][10] |        | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ )<br>MENDL-2p [15] |        | MENDL-2 [16] |        |
|---------|---------------------|---------------------------------|--------|-------------------------------------------------------------------------|--------|--------------|--------|
|         |                     | [11]                            | [12]   | [11]                                                                    | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 3835.72                         | 57.06  | 3835.72                                                                 | 57.06  | 3835.72      | 57.06  |
| Be-7    | 53.3                | 172.02                          | 1.41   | 172.02                                                                  | 1.41   | 172.02       | 1.41   |
| Be-10   | 5.84E+08            | 90.55                           | 0.74   | 90.55                                                                   | 0.74   | 90.55        | 0.74   |
| B-11    | 1.10E+10            | 4.66                            | 0.02   | 4.66                                                                    | 0.02   | 4.66         | 0.02   |
| C-14    | 2.09E+06            | 42.72                           | 0.22   | 42.72                                                                   | 0.22   | 42.72        | 0.22   |
| Ne-20   | 1.10E+10            | 4.16                            | 0.02   | 4.16                                                                    | 0.02   | 4.16         | 0.02   |
| Na-22   | 950.35              | 48.48                           | 0.26   | 48.48                                                                   | 0.26   | 48.48        | 0.26   |
| Al-26   | 2.66E+08            | 43.59                           | 0.27   | 43.59                                                                   | 0.27   | 43.59        | 0.27   |
| Si-32   | 237000              | 34.44                           | 0.28   | 34.44                                                                   | 0.28   | 34.44        | 0.28   |
| P-31    | 1.10E+10            | 10.39                           | 0.08   | 10.39                                                                   | 0.08   | 10.39        | 0.08   |
| P-32    | 14.28               | 213.35                          | 1.73   | 213.35                                                                  | 1.73   | 213.35       | 1.73   |
| P-33    | 25.3                | 125.60                          | 1.06   | 125.60                                                                  | 1.06   | 125.60       | 1.06   |
| S-35    | 87.4                | 205.59                          | 1.94   | 205.59                                                                  | 1.94   | 205.59       | 1.94   |
| Cl-36   | 1.10E+08            | 383.74                          | 3.84   | 383.74                                                                  | 3.84   | 383.74       | 3.84   |
| Ar-37   | 35                  | 190.88                          | 2.03   | 190.22                                                                  | 2.04   | 190.88       | 2.03   |
| Ar-39   | 98200               | 385.12                          | 4.67   | 343.95                                                                  | 3.67   | 385.12       | 4.66   |
| Ar-42   | 12000               | 14.60                           | 0.22   | 13.26                                                                   | 0.18   | 14.60        | 0.20   |
| K-42    | 0.52                | 220.29                          | 3.26   | 177.52                                                                  | 2.23   | 212.36       | 3.07   |
| K-43    | 0.93                | 83.72                           | 1.32   | 65.21                                                                   | 0.88   | 79.40        | 1.22   |
| Ca-40   | 1.10E+10            | 3.10                            | 0.04   | 7.74                                                                    | 0.16   | 3.10         | 0.04   |
| Ca-41   | 3.65E+07            | 220.03                          | 3.04   | 357.25                                                                  | 6.56   | 220.03       | 3.04   |
| Ca-45   | 165                 | 158.70                          | 2.82   | 113.35                                                                  | 1.73   | 154.95       | 2.71   |
| Ca-47   | 4.536               | 16.13                           | 0.32   | 10.83                                                                   | 0.19   | 14.39        | 0.27   |
| Sc-46   | 83.8                | 856.98                          | 16.10  | 637.57                                                                  | 10.87  | 811.84       | 15.02  |
| Sc-47   | 3.42                | 376.04                          | 7.43   | 255.64                                                                  | 4.52   | 397.65       | 7.91   |
| Sc-48   | 1.82                | 58.96                           | 1.22   | 48.89                                                                   | 0.98   | 92.91        | 2.02   |
| Ti-44   | 17200               | 56.49                           | 0.94   | 83.79                                                                   | 1.63   | 56.49        | 0.94   |
| V-48    | 15.976              | 1870.55                         | 38.65  | 2499.58                                                                 | 53.63  | 1987.74      | 41.32  |
| V-49    | 330                 | 6521.35                         | 141.46 | 5293.46                                                                 | 112.61 | 6012.03      | 129.29 |
| V-51    | 1.10E+10            | 164.48                          | 3.83   | 87.21                                                                   | 1.88   | 226.53       | 5.41   |
| Cr-48   | 0.989               | 77.93                           | 1.60   | 94.81                                                                   | 2.01   | 73.36        | 1.44   |
| Cr-51   | 27.7                | 11549.30                        | 269.01 | 8345.33                                                                 | 186.75 | 10777.50     | 246.21 |
| Mn-52   | 5.59                | 965.00                          | 22.79  | 4582.43                                                                 | 115.37 | 2843.82      | 72.47  |
| Mn-53   | 1.38E+09            | 2227.62                         | 54.15  | 5554.44                                                                 | 145.08 | 5387.28      | 141.47 |
| Mn-54   | 312                 | 12663.10                        | 299.66 | 15127.20                                                                | 368.94 | 26856.90     | 667.23 |
| Fe-55   | 986.2               | 6650.62                         | 207.97 | 6638.71                                                                 | 427.22 | 6650.62      | 207.97 |



Table 16.1: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in molybdenum by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |               |       |              |       |
|---------|---------------------|--------------------------------------------------------|-------|---------------|-------|--------------|-------|
|         |                     | Semi-empirical formulae [9][10]                        |       | MENDL-2p [15] |       | MENDL-2 [16] |       |
|         |                     | [11]                                                   | [12]  | [11]          | [12]  | [11]         | [12]  |
| H-3     | 4503.4              | 31.63                                                  | 37.28 | 31.63         | 37.28 | 31.63        | 37.28 |
| Be-7    | 53.3                | 0.65                                                   | 0.25  | 0.65          | 0.25  | 0.65         | 0.25  |
| Be-10   | 5.84E+08            | 4.08                                                   | 8.88  | 4.08          | 8.88  | 4.08         | 8.88  |
| B-11    | 1.10E+10            | 0.03                                                   | 0.01  | 0.03          | 0.01  | 0.03         | 0.01  |
| C-14    | 2.09E+06            | 0.45                                                   | 0.12  | 0.45          | 0.12  | 0.45         | 0.12  |
| Ne-20   | 1.10E+10            | 0.02                                                   | 0.00  | 0.02          | 0.00  | 0.02         | 0.00  |
| Na-22   | 950.35              | 0.11                                                   | 0.03  | 0.11          | 0.03  | 0.11         | 0.03  |
| Al-26   | 2.66E+08            | 0.07                                                   | 0.02  | 0.07          | 0.02  | 0.07         | 0.02  |
| Si-32   | 237000              | 0.03                                                   | 0.01  | 0.03          | 0.01  | 0.03         | 0.01  |
| P-31    | 1.10E+10            | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| P-32    | 14.28               | 0.20                                                   | 0.05  | 0.20          | 0.05  | 0.20         | 0.05  |
| P-33    | 25.3                | 0.11                                                   | 0.03  | 0.11          | 0.03  | 0.11         | 0.03  |
| S-35    | 87.4                | 0.12                                                   | 0.03  | 0.12          | 0.03  | 0.12         | 0.03  |
| Cl-36   | 1.10E+08            | 0.20                                                   | 0.05  | 0.20          | 0.05  | 0.20         | 0.05  |
| Ar-37   | 35                  | 0.10                                                   | 0.03  | 0.10          | 0.03  | 0.10         | 0.03  |
| Ar-39   | 98200               | 0.16                                                   | 0.04  | 0.16          | 0.04  | 0.16         | 0.04  |
| Ar-42   | 12000               | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| K-42    | 0.52                | 0.06                                                   | 0.02  | 0.06          | 0.02  | 0.06         | 0.02  |
| K-43    | 0.93                | 0.02                                                   | 0.01  | 0.02          | 0.01  | 0.02         | 0.01  |
| Ca-40   | 1.10E+10            | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ca-41   | 3.65E+07            | 0.07                                                   | 0.02  | 0.07          | 0.02  | 0.07         | 0.02  |
| Ca-45   | 165                 | 0.03                                                   | 0.01  | 0.03          | 0.01  | 0.03         | 0.01  |
| Ca-47   | 4.536               | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Sc-46   | 83.8                | 0.10                                                   | 0.03  | 0.10          | 0.03  | 0.10         | 0.03  |
| Sc-47   | 3.42                | 0.04                                                   | 0.01  | 0.04          | 0.01  | 0.04         | 0.01  |
| Sc-48   | 1.82                | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Ti-44   | 17200               | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| V-48    | 15.976              | 0.14                                                   | 0.05  | 0.14          | 0.05  | 0.14         | 0.05  |
| V-49    | 330                 | 0.36                                                   | 0.15  | 0.36          | 0.15  | 0.36         | 0.15  |
| V-51    | 1.10E+10            | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Cr-48   | 0.989               | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Cr-51   | 27.7                | 0.36                                                   | 0.16  | 0.36          | 0.16  | 0.36         | 0.16  |
| Mn-52   | 5.59                | 0.14                                                   | 0.06  | 0.14          | 0.06  | 0.14         | 0.06  |
| Mn-53   | 1.38E+09            | 0.16                                                   | 0.07  | 0.16          | 0.07  | 0.16         | 0.07  |
| Mn-54   | 312                 | 0.35                                                   | 0.16  | 0.35          | 0.16  | 0.35         | 0.16  |
| Fe-55   | 986.2               | 0.36                                                   | 0.17  | 0.36          | 0.17  | 0.36         | 0.17  |
| Fe-59   | 44.6                | 0.04                                                   | 0.02  | 0.04          | 0.02  | 0.04         | 0.02  |
| Fe-60   | 5.47E+08            | 0.01                                                   | 0.01  | 0.01          | 0.01  | 0.01         | 0.01  |
| Co-56   | 78.8                | 0.11                                                   | 0.05  | 0.11          | 0.05  | 0.11         | 0.05  |
| Co-57   | 271                 | 0.38                                                   | 0.19  | 0.38          | 0.19  | 0.38         | 0.19  |
| Co-58   | 70.8                | 0.49                                                   | 0.24  | 0.49          | 0.24  | 0.49         | 0.24  |
| Co-60   | 1925.2              | 0.16                                                   | 0.08  | 0.16          | 0.08  | 0.16         | 0.08  |
| Ni-56   | 6.1                 | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ni-57   | 1.5                 | 0.02                                                   | 0.01  | 0.02          | 0.01  | 0.02         | 0.01  |
| Ni-59   | 2.92E+07            | 0.34                                                   | 0.18  | 0.34          | 0.18  | 0.34         | 0.18  |
| Ni-63   | 36524               | 0.11                                                   | 0.06  | 0.11          | 0.06  | 0.11         | 0.06  |
| Ni-66   | 2.28                | 0.01                                                   | 0.00  | 0.01          | 0.00  | 0.01         | 0.00  |
| Cu-63   | 1.10E+10            | 0.05                                                   | 0.03  | 0.05          | 0.03  | 0.05         | 0.03  |
| Cu-67   | 2.58                | 0.02                                                   | 0.02  | 0.02          | 0.02  | 0.02         | 0.02  |
| Zn-65   | 244.1               | 1.22                                                   | 0.76  | 1.22          | 0.76  | 1.22         | 0.76  |
| Zn-72   | 1.938               | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Ga-67   | 3.263               | 1.44                                                   | 0.98  | 1.44          | 0.98  | 1.44         | 0.98  |
| Ga-69   | 1.10E+10            | 0.03                                                   | 0.02  | 0.03          | 0.02  | 0.03         | 0.02  |
| Ga-71   | 0.59                | 0.02                                                   | 0.02  | 0.02          | 0.02  | 0.02         | 0.02  |
| Ge-68   | 270.8               | 1.03                                                   | 0.75  | 1.03          | 0.75  | 1.03         | 0.75  |
| Ge-69   | 1.625               | 1.46                                                   | 1.08  | 1.46          | 1.08  | 1.46         | 1.08  |
| Ge-71   | 11.2                | 2.32                                                   | 1.84  | 2.32          | 1.84  | 2.32         | 1.84  |
| As-72   | 1.083               | 1.42                                                   | 1.14  | 1.42          | 1.14  | 1.42         | 1.14  |
| As-73   | 80.3                | 2.71                                                   | 2.33  | 2.71          | 2.33  | 2.71         | 2.33  |
| As-74   | 17.8                | 0.43                                                   | 0.36  | 0.43          | 0.36  | 0.43         | 0.36  |

Table 16.2: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in molybdenum by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae [9][10]                        |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| As-76   | 1.096               | 0.07                                                   | 0.06   | 0.07          | 0.06   | 0.07         | 0.06   |
| As-77   | 1.62                | 0.03                                                   | 0.03   | 0.03          | 0.03   | 0.03         | 0.03   |
| Se-72   | 8.4                 | 0.93                                                   | 0.81   | 0.93          | 0.81   | 0.93         | 0.81   |
| Se-75   | 118.5               | 3.26                                                   | 3.11   | 3.26          | 3.09   | 3.26         | 3.11   |
| Se-79   | 2.38E+07            | 0.05                                                   | 0.05   | 0.05          | 0.05   | 0.05         | 0.05   |
| Se-80   | 1.10E+10            | 0.00                                                   | 0.00   | 0.00          | 0.00   | 0.00         | 0.00   |
| Br-76   | 0.67                | 2.96                                                   | 3.01   | 2.96          | 3.03   | 2.96         | 3.01   |
| Br-77   | 2.375               | 3.99                                                   | 4.19   | 4.07          | 4.40   | 3.99         | 4.19   |
| Br-82   | 1.47                | 0.03                                                   | 0.04   | 0.03          | 0.04   | 0.03         | 0.04   |
| Kr-79   | 1.46                | 4.81                                                   | 5.67   | 5.11          | 6.46   | 4.81         | 5.67   |
| Kr-81   | 7.66E+07            | 7.26                                                   | 9.53   | 7.93          | 11.24  | 7.29         | 9.58   |
| Kr-85   | 3908.1              | 0.03                                                   | 0.04   | 0.03          | 0.04   | 0.03         | 0.04   |
| Rb-83   | 86.2                | 1.92                                                   | 2.60   | 1.94          | 2.66   | 1.93         | 2.62   |
| Rb-84   | 32.9                | 0.77                                                   | 1.09   | 0.77          | 1.10   | 0.78         | 1.12   |
| Rb-86   | 18.8                | 0.15                                                   | 0.24   | 0.14          | 0.21   | 0.15         | 0.25   |
| Sr-82   | 25                  | 6.09                                                   | 9.07   | 7.50          | 12.66  | 6.18         | 9.27   |
| Sr-83   | 1.35                | 7.87                                                   | 12.01  | 9.08          | 15.09  | 7.88         | 12.01  |
| Sr-85   | 64.8                | 14.42                                                  | 25.12  | 15.64         | 28.32  | 15.36        | 26.96  |
| Sr-89   | 50.5                | 0.09                                                   | 0.15   | 0.08          | 0.13   | 0.14         | 0.28   |
| Sr-90   | 10600               | 0.02                                                   | 0.04   | 0.02          | 0.03   | 0.05         | 0.11   |
| Y-86    | 0.61                | 6.95                                                   | 11.43  | 8.51          | 15.31  | 7.90         | 13.73  |
| Y-87    | 3.35                | 23.39                                                  | 45.82  | 25.93         | 52.23  | 25.37        | 50.28  |
| Y-88    | 106.6               | 2.27                                                   | 3.83   | 4.09          | 8.26   | 4.19         | 8.46   |
| Y-89    | 1.10E+10            | 0.07                                                   | 0.11   | 0.15          | 0.31   | 0.20         | 0.45   |
| Y-90    | 2.67                | 0.38                                                   | 0.66   | 0.53          | 1.02   | 1.15         | 2.50   |
| Y-91    | 58.5                | 0.11                                                   | 0.14   | 0.19          | 0.34   | 0.65         | 1.45   |
| Zr-86   | 0.688               | 3.44                                                   | 6.15   | 8.57          | 18.80  | 4.73         | 9.30   |
| Zr-88   | 83.4                | 32.19                                                  | 65.79  | 34.61         | 71.67  | 36.04        | 74.97  |
| Zr-89   | 3.27                | 37.01                                                  | 77.72  | 43.93         | 95.15  | 45.44        | 99.26  |
| Zr-92   | 1.10E+10            | 0.22                                                   | 0.40   | 0.20          | 0.35   | 0.43         | 0.96   |
| Zr-93   | 5.48E+08            | 1.27                                                   | 2.29   | 1.03          | 1.71   | 2.72         | 6.10   |
| Zr-95   | 64                  | 0.62                                                   | 1.09   | 0.59          | 1.02   | 2.39         | 5.64   |
| Nb-91   | 3.65E+06            | 55.11                                                  | 124.98 | 76.42         | 180.15 | 103.90       | 252.45 |
| Nb-91   | 62                  | 36.01                                                  | 79.98  | 41.63         | 94.82  | 58.29        | 137.52 |
| Nb-93   | 4967                | 38.40                                                  | 88.44  | 18.00         | 37.41  | 38.92        | 90.00  |
| Nb-94   | 7.30E+06            | 38.84                                                  | 92.83  | 17.87         | 39.99  | 33.87        | 78.25  |
| Nb-95   | 35                  | 34.65                                                  | 85.84  | 17.89         | 43.45  | 29.39        | 68.82  |
| Nb-96   | 0.975               | 28.74                                                  | 73.36  | 15.45         | 39.56  | 23.50        | 56.16  |
| Mo-93   | 1.10E+06            | 36.84                                                  | 90.83  | 67.93         | 170.99 | 72.09        | 184.37 |
| Mo-99   | 2.75                | 12.11                                                  | 28.13  | 5.95          | 12.18  | 19.92        | 48.25  |
| Tc-95   | 61                  | 35.79                                                  | 97.80  | 50.44         | 140.77 | 35.79        | 97.80  |
| Tc-96   | 4.3                 | 36.01                                                  | 97.82  | 37.80         | 107.80 | 36.01        | 97.82  |
| Tc-97   | 90                  | 27.80                                                  | 76.05  | 37.15         | 102.50 | 27.80        | 76.05  |
| Tc-97   | 9.49E+08            | 14.41                                                  | 39.41  | 19.26         | 53.12  | 14.41        | 39.41  |
| Tc-98   | 1.53E+09            | 6.34                                                   | 18.84  | 5.08          | 18.22  | 6.34         | 18.84  |
| Tc-99   | 8.39E+07            | 8.10                                                   | 21.54  | 9.98          | 26.60  | 8.10         | 21.54  |

Table 17: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in nitrogen by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life (days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |               |       |              |       |
|---------|------------------|--------------------------------------------------------|-------|---------------|-------|--------------|-------|
|         |                  | Semi-empirical formulae [9][10]                        |       | MENDL-2p [15] |       | MENDL-2 [16] |       |
|         |                  | [11]                                                   | [12]  | [11]          | [12]  | [11]         | [12]  |
| H-3     | 4503.4           | 0.00                                                   | 0.00  | 0.00          | 0.00  | 0.00         | 0.00  |
| Be-7    | 53.3             | 46.97                                                  | 99.47 | 46.97         | 99.47 | 46.97        | 99.47 |
| Be-10   | 5.84E+08         | 3.10                                                   | 5.14  | 3.10          | 5.14  | 3.10         | 5.14  |
| B-11    | 1.10E+10         | 8.17                                                   | 19.19 | 8.17          | 19.19 | 8.17         | 19.19 |
| C-14    | 2.09E+06         | 9.18                                                   | 26.40 | 9.18          | 26.40 | 9.18         | 26.40 |

Table 18: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in nickel by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life<br>(days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|---------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                     | Semi-empirical formulae [9][10]                        |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                     | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4              | 37.03                                                  | 53.47  | 37.03         | 53.47  | 37.03        | 53.47  |
| Be-7    | 53.3                | 2.16                                                   | 1.77   | 2.16          | 1.77   | 2.16         | 1.77   |
| Be-10   | 5.84E+08            | 0.45                                                   | 0.36   | 0.45          | 0.36   | 0.45         | 0.36   |
| B-11    | 1.10E+10            | 0.04                                                   | 0.02   | 0.04          | 0.02   | 0.04         | 0.02   |
| C-14    | 2.09E+06            | 0.33                                                   | 0.17   | 0.33          | 0.17   | 0.33         | 0.17   |
| Ne-20   | 1.10E+10            | 0.05                                                   | 0.02   | 0.05          | 0.02   | 0.05         | 0.02   |
| Na-22   | 950.35              | 0.51                                                   | 0.26   | 0.51          | 0.26   | 0.51         | 0.26   |
| Al-26   | 2.66E+08            | 0.47                                                   | 0.26   | 0.47          | 0.26   | 0.47         | 0.26   |
| Si-32   | 237000              | 0.14                                                   | 0.10   | 0.14          | 0.10   | 0.14         | 0.10   |
| P-31    | 1.10E+10            | 0.10                                                   | 0.06   | 0.10          | 0.06   | 0.10         | 0.06   |
| P-32    | 14.28               | 1.11                                                   | 0.75   | 1.11          | 0.75   | 1.11         | 0.75   |
| P-33    | 25.3                | 0.49                                                   | 0.35   | 0.49          | 0.35   | 0.49         | 0.35   |
| S-35    | 87.4                | 0.80                                                   | 0.62   | 0.80          | 0.62   | 0.80         | 0.62   |
| Cl-36   | 1.10E+08            | 2.02                                                   | 1.66   | 2.02          | 1.66   | 2.02         | 1.66   |
| Ar-37   | 35                  | 2.09                                                   | 1.84   | 2.09          | 1.84   | 2.09         | 1.84   |
| Ar-39   | 98200               | 1.39                                                   | 1.33   | 1.39          | 1.33   | 1.39         | 1.33   |
| Ar-42   | 12000               | 0.04                                                   | 0.04   | 0.04          | 0.04   | 0.04         | 0.04   |
| K-42    | 0.52                | 0.74                                                   | 0.86   | 0.74          | 0.86   | 0.74         | 0.86   |
| K-43    | 0.93                | 0.26                                                   | 0.32   | 0.26          | 0.32   | 0.26         | 0.32   |
| Ca-40   | 1.10E+10            | 0.07                                                   | 0.07   | 0.07          | 0.07   | 0.07         | 0.07   |
| Ca-41   | 3.65E+07            | 3.27                                                   | 3.68   | 3.26          | 3.67   | 3.24         | 3.61   |
| Ca-45   | 165                 | 0.44                                                   | 0.62   | 0.44          | 0.62   | 0.43         | 0.59   |
| Ca-47   | 4.536               | 0.03                                                   | 0.05   | 0.03          | 0.05   | 0.03         | 0.05   |
| Sc-46   | 83.8                | 2.32                                                   | 3.52   | 1.93          | 2.58   | 2.20         | 3.24   |
| Sc-47   | 3.42                | 0.90                                                   | 1.45   | 0.70          | 0.97   | 0.86         | 1.35   |
| Sc-48   | 1.82                | 0.12                                                   | 0.20   | 0.10          | 0.16   | 0.11         | 0.19   |
| Ti-44   | 17200               | 1.32                                                   | 1.82   | 1.39          | 2.03   | 1.29         | 1.75   |
| V-48    | 15.976              | 13.22                                                  | 23.25  | 13.47         | 24.06  | 12.94        | 22.55  |
| V-49    | 330                 | 21.83                                                  | 40.14  | 21.64         | 40.18  | 20.51        | 36.98  |
| V-51    | 1.10E+10            | 0.20                                                   | 0.40   | 0.12          | 0.21   | 0.18         | 0.35   |
| Cr-48   | 0.989               | 1.31                                                   | 2.32   | 1.67          | 3.21   | 1.23         | 2.14   |
| Cr-51   | 27.7                | 36.61                                                  | 74.53  | 34.67         | 69.66  | 36.84        | 74.71  |
| Mn-52   | 5.59                | 30.56                                                  | 65.81  | 31.02         | 67.49  | 31.92        | 69.39  |
| Mn-53   | 1.38E+09            | 30.08                                                  | 66.93  | 23.63         | 51.23  | 29.22        | 65.14  |
| Mn-54   | 312                 | 39.33                                                  | 89.93  | 24.26         | 52.64  | 38.47        | 87.61  |
| Fe-55   | 986.2               | 203.72                                                 | 484.69 | 126.79        | 293.37 | 175.67       | 414.42 |
| Fe-59   | 44.6                | 0.82                                                   | 1.83   | 0.44          | 0.89   | 1.48         | 3.58   |
| Fe-60   | 5.47E+08            | 0.15                                                   | 0.30   | 0.10          | 0.19   | 0.23         | 0.52   |
| Co-56   | 78.8                | 85.49                                                  | 204.65 | 90.04         | 223.31 | 101.71       | 252.70 |
| Co-57   | 271                 | 137.93                                                 | 315.32 | 171.25        | 404.74 | 256.06       | 624.33 |
| Co-58   | 70.8                | 46.30                                                  | 121.44 | 36.38         | 97.41  | 87.44        | 491.91 |
| Co-60   | 1925.2              | 16.38                                                  | 47.73  | 13.66         | 40.84  | 12.94        | 35.96  |
| Ni-56   | 6.1                 | 7.29                                                   | 17.49  | 17.27         | 44.08  | 7.16         | 17.55  |
| Ni-57   | 1.5                 | 64.73                                                  | 153.14 | 100.37        | 249.43 | 61.52        | 144.92 |
| Ni-59   | 2.92E+07            | 33.45                                                  | 79.71  | 55.67         | 139.16 | 40.45        | 97.79  |
| Ni-63   | 36524               | 1.84                                                   | 4.27   | 0.96          | 2.00   | 2.43         | 5.78   |
| Ni-66   | 2.28                | 0.00                                                   | 0.13   | 0.00          | 0.00   | 0.00         | 0.00   |
| Cu-63   | 1.10E+10            | 0.05                                                   | 0.01   | 0.08          | 0.22   | 0.05         | 0.13   |

Table 19: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in phosphorus by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life (days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |       |               |       |              |       |
|---------|------------------|--------------------------------------------------------|-------|---------------|-------|--------------|-------|
|         |                  | Semi-empirical formulae [9][10]                        |       | MENDL-2p [15] |       | MENDL-2 [16] |       |
|         |                  | [11]                                                   | [12]  | [11]          | [12]  | [11]         | [12]  |
| H-3     | 4503.4           | 5326                                                   | 90.87 | 5326          | 90.87 | 5326         | 90.87 |
| Be-7    | 53.3             | 450.00                                                 | 4.91  | 450.00        | 4.91  | 450.00       | 4.91  |
| Be-10   | 5.84E+08         | 75.24                                                  | 0.65  | 75.24         | 0.65  | 75.24        | 0.65  |
| B-11    | 1.10E+10         | 33.77                                                  | 0.45  | 33.77         | 0.45  | 33.77        | 0.45  |
| C-14    | 2.09E+06         | 196.17                                                 | 2.48  | 196.17        | 2.48  | 196.17       | 2.48  |
| Ne-20   | 1.10E+10         | 101.96                                                 | 1.85  | 114.46        | 2.14  | 103.69       | 1.87  |
| Na-22   | 950.35           | 1900                                                   | 36.01 | 2297          | 45.80 | 1965         | 37.43 |
| Al-26   | 2.66E+08         | 2639                                                   | 55.90 | 3852          | 87.32 | 3506         | 78.37 |

Table 20: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in sulphur by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life (days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                  | Semi-empirical formulae [9][10]                        |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                  | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4           | 51.74                                                  | 87.40  | 51.74         | 87.40  | 51.74        | 87.40  |
| Be-7    | 53.3             | 4.50                                                   | 4.78   | 4.50          | 4.78   | 4.50         | 4.78   |
| Be-10   | 5.84E+08         | 0.54                                                   | 0.42   | 0.54          | 0.42   | 0.54         | 0.42   |
| B-11    | 1.10E+10         | 0.28                                                   | 0.36   | 0.28          | 0.36   | 0.28         | 0.36   |
| C-14    | 2.09E+06         | 1.43                                                   | 1.67   | 1.43          | 1.67   | 1.43         | 1.67   |
| Ne-20   | 1.10E+10         | 0.99                                                   | 1.73   | 0.96          | 1.67   | 0.94         | 1.73   |
| Na-22   | 950.35           | 18.90                                                  | 34.30  | 20.02         | 36.94  | 19.30        | 34.30  |
| Al-26   | 2.66E+08         | 27.55                                                  | 56.59  | 34.03         | 72.31  | 33.17        | 56.59  |
| Si-32   | 237000           | 0.76                                                   | 1.14   | 0.68          | 0.94   | 1.00         | 1.14   |
| P-31    | 1.10E+10         | 6.49                                                   | 13.98  | 14.06         | 34.00  | 20.84        | 13.98  |
| P-32    | 14.28            | 10.98                                                  | 30.88  | 11.00         | 31.10  | 75.92        | 30.88  |
| P-33    | 25.3             | 2.98                                                   | 6.61   | 2.40          | 5.12   | 7.71         | 6.61   |
| S-35    | 87.4             | 0.0427                                                 | 0.0992 | 0.0291        | 0.0646 | 0.0858       | 0.0992 |
| Cl-36   | 1.10E+08         | 0.02                                                   | 0.06   | 0.02          | 0.19   | 0.02         | 0.06   |

Table 21: Calculations of production rates (in units of  $\text{kg}^{-1} \text{ day}^{-1}$  of radioactive isotopes in silicon by ACTIVIA. See Table 8 for details of the table.

| Isotope | Half-life (days) | Production rates ( $\text{kg}^{-1} \text{ day}^{-1}$ ) |        |               |        |              |        |
|---------|------------------|--------------------------------------------------------|--------|---------------|--------|--------------|--------|
|         |                  | Semi-empirical formulae [9][10]                        |        | MENDL-2p [15] |        | MENDL-2 [16] |        |
|         |                  | [11]                                                   | [12]   | [11]          | [12]   | [11]         | [12]   |
| H-3     | 4503.4           | 56.54                                                  | 97.83  | 56.54         | 97.83  | 56.54        | 97.83  |
| Be-7    | 53.3             | 6.29                                                   | 8.09   | 6.29          | 8.09   | 6.29         | 8.09   |
| Be-10   | 5.84E+08         | 0.71                                                   | 0.66   | 0.71          | 0.66   | 0.71         | 0.66   |
| B-11    | 1.10E+10         | 0.43                                                   | 0.63   | 0.43          | 0.63   | 0.43         | 0.63   |
| C-14    | 2.09E+06         | 2.13                                                   | 2.87   | 2.13          | 2.87   | 2.09         | 2.79   |
| Ne-20   | 1.10E+10         | 2.01                                                   | 4.11   | 2.62          | 5.65   | 2.57         | 5.53   |
| Na-22   | 950.35           | 40.73                                                  | 86.06  | 57.89         | 127.12 | 57.17        | 125.74 |
| Al-26   | 2.66E+08         | 119.85                                                 | 284.96 | 114.51        | 274.26 | 136.04       | 330.28 |